10 minutes to pandas
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Indexing and selecting data

The axis labeling information in pandas objects serves many purposes:

- Identifies data (i.e. provides *metadata*) using known indicators, important for analysis, visualization, and interactive console display.
- Enables automatic and explicit data alignment.
- Allows intuitive getting and setting of subsets of the data set.

In this section, we will focus on the final point: namely, how to slice, dice, and generally get and set subsets of pandas objects. The primary focus will be on Series and DataFrame as they have received more development attention in this area.

> **Note**
>
> The Python and NumPy indexing operators `[]` and attribute operator `.` provide quick and easy access to pandas data structures across a wide range of use cases. This makes interactive work intuitive, as there's little new to learn if you already know how to deal with Python dictionaries and NumPy arrays. However, since the type of the data to be accessed isn't known in advance, directly using standard operators has some optimization limits. For production code, we recommended that you take advantage of the optimized pandas data access methods exposed in this chapter.

See the MultiIndex / Advanced Indexing for `MultiIndex` and more advanced indexing documentation.

See the cookbook for some advanced strategies.

Different choices for indexing

Object selection has had a number of user-requested additions in order to support more explicit location based indexing. pandas now supports three types of multi-axis indexing.

- `.loc` is primarily label based, but may also be used with a boolean array. `.loc` will raise `KeyError` when the items are not found. Allowed inputs are:
  - A single label, e.g. `5` or `'a'` (Note that `5` is interpreted as a *label* of the index. This use is **not** an integer position along the index.).
  - A list or array of labels `['a', 'b', 'c']`.
  - A slice object with labels `'a':'f'` (Note that contrary to usual Python slices, **both** the start and the stop are included, when present in the index! See Slicing with labels and Endpoints are inclusive.)
  - A boolean array (any `NA` values will be treated as `False`).
  - A `callable` function with one argument (the calling Series or DataFrame) and that returns valid output for indexing (one of the above).
  - A tuple of row (and column) indices whose elements are one of the above inputs.

  See more at Selection by Label.

- `.iloc` is primarily integer position based (from `0` to `length-1` of the axis), but may also be used with a boolean array. `.iloc` will raise `IndexError` if a requested indexer is out-of-bounds, except *slice* indexers which allow out-of-bounds indexing. (this conforms with Python/NumPy *slice* semantics). Allowed inputs are:
  - An integer e.g. `5`.
  - A list or array of integers `[4, 3, 0]`.
  - A slice object with ints `1:7`.
  - A boolean array (any `NA` values will be treated as `False`).
  - A `callable` function with one argument (the calling Series or DataFrame) and that returns valid output for indexing (one of the above).
  - A tuple of row (and column) indices whose elements are one of the above inputs.

  See more at Selection by Position, Advanced Indexing and Advanced Hierarchical.

- `.loc`, `.iloc`, and also `[]` indexing can accept a `callable` as indexer. See more at Selection By Callable.

> **Note**
>
> Destructuring tuple keys into row (and column) indexes occurs *before* callables are applied, so you cannot return a tuple from a callable to index both rows and columns.

Getting values from an object with multi-axes selection uses the following notation (using `.loc` as an example, but the following applies to `.iloc` as well). Any of the axes accessors may be the null slice `:`. Axes left out of the specification are assumed to be `:`, e.g. `p.loc['a']` is equivalent to `p.loc['a', :]`.

```
In [1]: ser = pd.Series(range(5), index=list("abcde"))

In [2]: ser.loc[["a", "c", "e"]]
Out[2]:
a    0
c    2
e    4
dtype: int64

In [3]: df = pd.DataFrame(np.arange(25).reshape(5, 5), index=list("abcde"), columns

In [4]: df.loc[["a", "c", "e"], ["b", "d"]]
Out[4]:
    b   d
a   1   3
c  11  13
e  21  23
```

Basics

As mentioned when introducing the data structures in the last section, the primary function of indexing with `[]` (a.k.a. `__getitem__` for those familiar with implementing class behavior in Python) is selecting out lower-dimensional slices. The following table shows return type values when indexing pandas objects with `[]`:

Object Type	Selection	Return Value Type
Series	`series[label]`	scalar value
DataFrame	`frame[colname]`	`Series` corresponding to colname

Here we construct a simple time series data set to use for illustrating the indexing functionality:

```
In [5]: dates = pd.date_range('1/1/2000', periods=8)

In [6]: df = pd.DataFrame(np.random.randn(8, 4),
   ...:                   index=dates, columns=['A', 'B', 'C', 'D'])
   ...:

In [7]: df
Out[7]:
                   A         B         C         D
2000-01-01  0.469112 -0.282863 -1.509059 -1.135632
2000-01-02  1.212112 -0.173215  0.119209 -1.044236
2000-01-03 -0.861849 -2.104569 -0.494929  1.071804
2000-01-04  0.721555 -0.706771 -1.039575  0.271860
2000-01-05 -0.424972  0.567020  0.276232 -1.087401
2000-01-06 -0.673690  0.113648 -1.478427  0.524988
2000-01-07  0.404705  0.577046 -1.715002 -1.039268
2000-01-08 -0.370647 -1.157892 -1.344312  0.844885
```

> **Note**
>
> None of the indexing functionality is time series specific unless specifically stated.

Thus, as per above, we have the most basic indexing using `[]`:

```
In [8]: s = df['A']

In [9]: s[dates[5]]
Out[9]: np.float64(-0.6736897080883706)
```

You can pass a list of columns to `[]` to select columns in that order. If a column is not contained in the DataFrame, an exception will be raised. Multiple columns can also be set in this manner:

```
In [10]: df
Out[10]:
                   A         B         C         D
2000-01-01  0.469112 -0.282863 -1.509059 -1.135632
2000-01-02  1.212112 -0.173215  0.119209 -1.044236
2000-01-03 -0.861849 -2.104569 -0.494929  1.071804
2000-01-04  0.721555 -0.706771 -1.039575  0.271860
2000-01-05 -0.424972  0.567020  0.276232 -1.087401
2000-01-06 -0.673690  0.113648 -1.478427  0.524988
2000-01-07  0.404705  0.577046 -1.715002 -1.039268
2000-01-08 -0.370647 -1.157892 -1.344312  0.844885

In [11]: df[['B', 'A']] = df[['A', 'B']]

In [12]: df
Out[12]:
                   A         B         C         D
2000-01-01 -0.282863  0.469112 -1.509059 -1.135632
2000-01-02 -0.173215  1.212112  0.119209 -1.044236
2000-01-03 -2.104569 -0.861849 -0.494929  1.071804
2000-01-04 -0.706771  0.721555 -1.039575  0.271860
2000-01-05  0.567020 -0.424972  0.276232 -1.087401
2000-01-06  0.113648 -0.673690 -1.478427  0.524988
2000-01-07  0.577046  0.404705 -1.715002 -1.039268
2000-01-08 -1.157892 -0.370647 -1.344312  0.844885
```

You may find this useful for applying a transform (in-place) to a subset of the columns.

> **Warning**
>
> pandas aligns all AXES when setting `Series` and `DataFrame` from `.loc`.
>
> This will **not** modify `df` because the column alignment is before value assignment.
>
> ```
> In [13]: df[['A', 'B']]
> Out[13]:
>                    A         B
> 2000-01-01 -0.282863  0.469112
> 2000-01-02 -0.173215  1.212112
> 2000-01-03 -2.104569 -0.861849
> 2000-01-04 -0.706771  0.721555
> 2000-01-05  0.567020 -0.424972
> 2000-01-06  0.113648 -0.673690
> 2000-01-07  0.577046  0.404705
> 2000-01-08 -1.157892 -0.370647
>
> In [14]: df.loc[:, ['B', 'A']] = df[['A', 'B']]
>
> In [15]: df[['A', 'B']]
> Out[15]:
>                    A         B
> 2000-01-01 -0.282863  0.469112
> 2000-01-02 -0.173215  1.212112
> 2000-01-03 -2.104569 -0.861849
> 2000-01-04 -0.706771  0.721555
> 2000-01-05  0.567020 -0.424972
> 2000-01-06  0.113648 -0.673690
> 2000-01-07  0.577046  0.404705
> 2000-01-08 -1.157892 -0.370647
> ```
>
> The correct way to swap column values is by using raw values:
>
> ```
> In [16]: df.loc[:, ['B', 'A']] = df[['A', 'B']].to_numpy()
>
> In [17]: df[['A', 'B']]
> Out[17]:
>                    A         B
> 2000-01-01  0.469112 -0.282863
> 2000-01-02  1.212112 -0.173215
> 2000-01-03 -0.861849 -2.104569
> 2000-01-04  0.721555 -0.706771
> 2000-01-05 -0.424972  0.567020
> 2000-01-06 -0.673690  0.113648
> 2000-01-07  0.404705  0.577046
> 2000-01-08 -0.370647 -1.157892
> ```
>
> However, pandas does not align AXES when setting `Series` and `DataFrame` from `.iloc` because `.iloc` operates by position.
>
> This will modify `df` because the column alignment is not done before value assignment.
>
> ```
> In [18]: df[['A', 'B']]
> Out[18]:
>                    A         B
> 2000-01-01  0.469112 -0.282863
> 2000-01-02  1.212112 -0.173215
> 2000-01-03 -0.861849 -2.104569
> 2000-01-04  0.721555 -0.706771
> 2000-01-05 -0.424972  0.567020
> 2000-01-06 -0.673690  0.113648
> 2000-01-07  0.404705  0.577046
> 2000-01-08 -0.370647 -1.157892
>
> In [19]: df.iloc[:, [1, 0]] = df[['A', 'B']]
>
> In [20]: df[['A','B']]
> Out[20]:
>                    A         B
> 2000-01-01 -0.282863  0.469112
> 2000-01-02 -0.173215  1.212112
> 2000-01-03 -2.104569 -0.861849
> 2000-01-04 -0.706771  0.721555
> 2000-01-05  0.567020 -0.424972
> 2000-01-06  0.113648 -0.673690
> 2000-01-07  0.577046  0.404705
> 2000-01-08 -1.157892 -0.370647
> ```

Attribute access

You may access an index on a `Series` or column on a `DataFrame` directly as an attribute:

```
In [21]: sa = pd.Series([1, 2, 3], index=list('abc'))

In [22]: dfa = df.copy()
```

```
In [23]: sa.b
Out[23]: np.int64(2)

In [24]: dfa.A
Out[24]:
2000-01-01   -0.282863
2000-01-02   -0.173215
2000-01-03   -2.104569
2000-01-04   -0.706771
2000-01-05    0.567020
2000-01-06    0.113648
2000-01-07    0.577046
2000-01-08   -1.157892
Freq: D, Name: A, dtype: float64
```

```
In [25]: sa.a = 5

In [26]: sa
Out[26]:
a    5
b    2
c    3
dtype: int64

In [27]: dfa.A = list(range(len(dfa.index)))  # ok if A already exists

In [28]: dfa
Out[28]:
            A         B         C         D
2000-01-01  0  0.469112 -1.509059 -1.135632
2000-01-02  1  1.212112  0.119209 -1.044236
2000-01-03  2 -0.861849 -0.494929  1.071804
2000-01-04  3  0.721555 -1.039575  0.271860
2000-01-05  4 -0.424972  0.276232 -1.087401
2000-01-06  5 -0.673690 -1.478427  0.524988
2000-01-07  6  0.404705 -1.715002 -1.039268
2000-01-08  7 -0.370647 -1.344312  0.844885

In [29]: dfa['A'] = list(range(len(dfa.index)))  # use this form to create a new co

In [30]: dfa
Out[30]:
            A         B         C         D
2000-01-01  0  0.469112 -1.509059 -1.135632
2000-01-02  1  1.212112  0.119209 -1.044236
2000-01-03  2 -0.861849 -0.494929  1.071804
2000-01-04  3  0.721555 -1.039575  0.271860
2000-01-05  4 -0.424972  0.276232 -1.087401
2000-01-06  5 -0.673690 -1.478427  0.524988
2000-01-07  6  0.404705 -1.715002 -1.039268
2000-01-08  7 -0.370647 -1.344312  0.844885
```

> **Warning**
>
> - You can use this access only if the index element is a valid Python identifier, e.g. `s.1` is not allowed. See here for an explanation of valid identifiers.
> - The attribute will not be available if it conflicts with an existing method name, e.g. `s.min` is not allowed, but `s['min']` is possible.
> - Similarly, the attribute will not be available if it conflicts with any of the following list: `index`, `major_axis`, `minor_axis`, `items`.
> - In any of these cases, standard indexing will still work, e.g. `s['1']`, `s['min']`, and `s['index']` will access the corresponding element or column.

If you are using the IPython environment, you may also use tab-completion to see these accessible attributes.

You can also assign a `dict` to a row of a `DataFrame`:

```
In [31]: x = pd.DataFrame({'x': [1, 2, 3], 'y': [3, 4, 5]})

In [32]: x.iloc[1] = {'x': 9, 'y': 99}

In [33]: x
Out[33]:
   x   y
0  1   3
1  9  99
2  3   5
```

You can use attribute access to modify an existing element of a Series or column of a DataFrame, but be careful; if you try to use attribute access to create a new column, it creates a new attribute rather than a new column and will this raise a `UserWarning`:

```
In [34]: df_new = pd.DataFrame({'one': [1., 2., 3.]})

In [35]: df_new.two = [4, 5, 6]

In [36]: df_new
Out[36]:
   one
0  1.0
1  2.0
2  3.0
```

Slicing ranges

The most robust and consistent way of slicing ranges along arbitrary axes is described in the Selection by Position section detailing the `.iloc` method. For now, we explain the semantics of slicing using the `[]` operator.

> **Note**
>
> When the `Series` has float indices, slicing will select by position.

With Series, the syntax works exactly as with an ndarray, returning a slice of the values and the corresponding labels:

```
In [37]: s[:5]
Out[37]:
2000-01-01    0.469112
2000-01-02    1.212112
2000-01-03   -0.861849
2000-01-04    0.721555
2000-01-05   -0.424972
Freq: D, Name: A, dtype: float64

In [38]: s[::2]
Out[38]:
2000-01-01    0.469112
2000-01-03   -0.861849
2000-01-05   -0.424972
2000-01-07    0.404705
Freq: 2D, Name: A, dtype: float64

In [39]: s[::-1]
Out[39]:
2000-01-08   -0.370647
2000-01-07    0.404705
2000-01-06   -0.673690
2000-01-05   -0.424972
2000-01-04    0.721555
2000-01-03   -0.861849
2000-01-02    1.212112
2000-01-01    0.469112
Freq: -1D, Name: A, dtype: float64
```

Note that setting works as well:

```
In [40]: s2 = s.copy()

In [41]: s2[:5] = 0

In [42]: s2
Out[42]:
2000-01-01    0.000000
2000-01-02    0.000000
2000-01-03    0.000000
2000-01-04    0.000000
2000-01-05    0.000000
2000-01-06   -0.673690
2000-01-07    0.404705
2000-01-08   -0.370647
Freq: D, Name: A, dtype: float64
```

With DataFrame, slicing inside of `[]` **slices the rows**. This is provided largely as a convenience since it is such a common operation.

```
In [43]: df[:3]
Out[43]:
                   A         B         C         D
2000-01-01 -0.282863  0.469112 -1.509059 -1.135632
2000-01-02 -0.173215  1.212112  0.119209 -1.044236
2000-01-03 -2.104569 -0.861849 -0.494929  1.071804

In [44]: df[::-1]
Out[44]:
                   A         B         C         D
2000-01-08 -1.157892 -0.370647 -1.344312  0.844885
2000-01-07  0.577046  0.404705 -1.715002 -1.039268
2000-01-06  0.113648 -0.673690 -1.478427  0.524988
2000-01-05  0.567020 -0.424972  0.276232 -1.087401
2000-01-04 -0.706771  0.721555 -1.039575  0.271860
2000-01-03 -2.104569 -0.861849 -0.494929  1.071804
2000-01-02 -0.173215  1.212112  0.119209 -1.044236
2000-01-01 -0.282863  0.469112 -1.509059 -1.135632
```

Selection by label

> **Warning**
>
> `.loc` is strict when you present slicers that are not compatible (or convertible) with the index type. For example using integers in a `DatetimeIndex`. These will raise a `TypeError`.
>
> ```
> In [45]: dfl = pd.DataFrame(np.random.randn(5, 4),
>    ....:                    columns=list('ABCD'),
>    ....:                    index=pd.date_range('20130101', periods=5))
>    ....:
>
> In [46]: dfl
> Out[46]:
>                    A         B         C         D
> 2013-01-01  1.075770 -0.109050  1.643563 -1.469388
> 2013-01-02  0.357021 -0.674600 -1.776904 -0.968914
> 2013-01-03 -1.294524  0.413738  0.276662 -0.472035
> 2013-01-04 -0.013960 -0.362543 -0.006154 -0.923061
> 2013-01-05  0.895717  0.805244 -1.206412  2.565646
>
> In [47]: dfl.loc[2:3]
> ---------------------------------------------------------------------------
> TypeError                                 Traceback (most recent call l
> Cell In[47], line 1
> ----> 1 dfl.loc[2:3]
>
> File ~/work/pandas/pandas/pandas/core/indexing.py:1207, in _LocationInd
>    1205 maybe_callable = com.apply_if_callable(key, self.obj)
>    1206 maybe_callable = self._raise_callable_usage(key, maybe_callable
> -> 1207 return self._getitem_axis(maybe_callable, axis=axis)
>
> File ~/work/pandas/pandas/pandas/core/indexing.py:1429, in _LocIndexer.
>    1427 if isinstance(key, slice):
>    1428     self._validate_key(key, axis)
> -> 1429     return self._get_slice_axis(key, axis=axis)
>    1430 elif com.is_bool_indexer(key):
>    1431     return self._getbool_axis(key, axis=axis)
>
> File ~/work/pandas/pandas/pandas/core/indexing.py:1461, in _LocIndexer.
>    1458     return obj.copy(deep=False)
>    1460 labels = obj._get_axis(axis)
> -> 1461 indexer = labels.slice_indexer(slice_obj.start, slice_obj.stop,
>    1463 if isinstance(indexer, slice):
>    1464     return self.obj._slice(indexer, axis=axis)
>
> File ~/work/pandas/pandas/pandas/core/indexes/datetimes.py:1072, in Dat
>    1064 # GH#33146 if start and end are combinations of str and None an
>    1065 # monotonic, we can not use Index.slice_indexer because it does
>    1066 # actual elements, is only searching for start and end
>    1067 if (
>    1068     check_str_or_none(start)
>    1069     or check_str_or_none(end)
>    1070     or self.is_monotonic_increasing
>    1071 ):
> -> 1072     return Index.slice_indexer(self, start, end, step)
>    1074 mask = np.array(True)
>    1075 in_index = True
>
> File ~/work/pandas/pandas/pandas/core/indexes/base.py:6804, in Index.sl
>    6753 def slice_indexer(
>    6754     self,
>    6755     start: Hashable | None = None,
>    6756     end: Hashable | None = None,
>    6757     step: int | None = None,
>    6758 ) -> slice:
>    6759     """
>    6760     Compute the slice indexer for input labels and step.
>    6761
>    (...)   6802     slice(1, 3, None)
>    6803     """
> -> 6804     start_slice, end_slice = self.slice_locs(start, end, step=s
>    6806     # return a slice
>    6807     if not is_scalar(start_slice):
>
> File ~/work/pandas/pandas/pandas/core/indexes/base.py:7062, in Index.sl
>    7060 start_slice = None
>    7061 if start is not None:
> -> 7062     start_slice = self.get_slice_bound(start, "left")
>    7063 if start_slice is None:
>    7064     start_slice = 0
>
> File ~/work/pandas/pandas/pandas/core/indexes/base.py:6964, in Index.ge
>    6960 original_label = label
>    6962 # For datetime indices label may be a string that has to be con
>    6963 # to datetime boundary according to its resolution.
> -> 6964 label = self._maybe_cast_slice_bound(label, side)
>    6966 # we need to look up the label
>    6967 try:
>
> File ~/work/pandas/pandas/pandas/core/indexes/datetimes.py:1032, in Dat
>    1023     label = Timestamp(label).to_pydatetime()
>    1024     warnings.warn(
>    1025         # GH#35830 deprecate last remaining inconsistent date t
>    1026         "Slicing with a datetime.date object is deprecated. "
>    (...)   1029         stacklevel=find_stack_level(),
>    1030     )
> -> 1032 label = super()._maybe_cast_slice_bound(label, side)
>    1033 self._data._assert_tzawareness_compat(label)
>    1034 return Timestamp(label)
>
> File ~/work/pandas/pandas/pandas/core/indexes/datetimelike.py:497, in D
>     495     return lower if side == "left" else upper
>     496 elif not isinstance(label, self._data._recognized_scalars):
> --> 497     self._raise_invalid_indexer("slice", label)
>     499 return label
>
> File ~/work/pandas/pandas/pandas/core/indexes/base.py:4125, in Index._r
>    4123 if reraise is not lib.no_default:
>    4124     raise TypeError(msg) from reraise
> -> 4125 raise TypeError(msg)
>
> TypeError: cannot do slice indexing on DatetimeIndex with these indexer
> ```
>
> String likes in slicing *can* be convertible to the type of the index and lead to natural slicing.
>
> ```
> In [48]: dfl.loc['20130102':'20130104']
> Out[48]:
>                    A         B         C         D
> 2013-01-02  0.357021 -0.674600 -1.776904 -0.968914
> 2013-01-03 -1.294524  0.413738  0.276662 -0.472035
> 2013-01-04 -0.013960 -0.362543 -0.006154 -0.923061
> ```

pandas provides a suite of methods in order to have **purely label based indexing**. This is a strict inclusion based protocol. Every label asked for must be in the index, or a `KeyError` will be raised. When slicing, both the start bound **AND** the stop bound are *included*, if present in the index. Integers are valid labels, but they refer to the label **and not the position**.

The `.loc` attribute is the primary access method. The following are valid inputs:

- A single label, e.g. `5` or `'a'` (Note that `5` is interpreted as a *label* of the index. This use is **not** an integer position along the index.).
- A list or array of labels `['a', 'b', 'c']`.
- A slice object with labels `'a':'f'`. Note that contrary to usual Python slices, **both** the start and the stop are included, when present in the index! See Slicing with labels.
- A boolean array.
- A `callable`, see Selection By Callable.

```
In [49]: s1 = pd.Series(np.random.randn(6), index=list('abcdef'))

In [50]: s1
Out[50]:
a    1.431256
b    1.340309
c   -1.170299
d   -0.226169
e    0.410835
f    0.813850
dtype: float64

In [51]: s1.loc['c':]
Out[51]:
c   -1.170299
d   -0.226169
e    0.410835
f    0.813850
dtype: float64

In [52]: s1.loc['b']
Out[52]: np.float64(1.3403088497993827)
```

Note that setting works as well:

```
In [53]: s1.loc['c':] = 0

In [54]: s1
Out[54]:
a    1.431256
b    1.340309
c    0.000000
d    0.000000
e    0.000000
f    0.000000
dtype: float64
```

With a DataFrame:

```
In [55]: df1 = pd.DataFrame(np.random.randn(6, 4),
   ....:                    index=list('abcdef'),
   ....:                    columns=list('ABCD'))
   ....:

In [56]: df1
Out[56]:
          A         B         C         D
a  0.132003 -0.827317 -0.076467 -1.187678
b  1.130127 -1.436737 -1.413681  1.607920
c  1.024180  0.569605  0.875906 -2.211372
d  0.974466 -2.006747 -0.410001 -0.078638
e  0.545952 -1.219217 -1.226825  0.769804
f -1.281247 -0.727707 -0.121306 -0.097883

In [57]: df1.loc[['a', 'b', 'd'], :]
Out[57]:
          A         B         C         D
a  0.132003 -0.827317 -0.076467 -1.187678
b  1.130127 -1.436737 -1.413681  1.607920
d  0.974466 -2.006747 -0.410001 -0.078638
```

Accessing via label slices:

```
In [58]: df1.loc['d':, 'A':'C']
Out[58]:
          A         B         C
d  0.974466 -2.006747 -0.410001
e  0.545952 -1.219217 -1.226825
f -1.281247 -0.727707 -0.121306
```

For getting a cross section using a label (equivalent to `df.xs('a')`):

```
In [59]: df1.loc['a']
Out[59]:
A    0.132003
B   -0.827317
C   -0.076467
D   -1.187678
Name: a, dtype: float64
```

For getting values with a boolean array:

```
In [60]: df1.loc['a'] > 0
Out[60]:
A     True
B    False
C    False
D    False
Name: a, dtype: bool

In [61]: df1.loc[:, df1.loc['a'] > 0]
Out[61]:
          A
a  0.132003
b  1.130127
c  1.024180
d  0.974466
e  0.545952
f -1.281247
```

NA values in a boolean array propagate as `False`:

```
In [62]: mask = pd.array([True, False, True, False, pd.NA, False], dtype="boolean")

In [63]: mask
Out[63]:
<BooleanArray>
```

```
[True, False, True, False, <NA>, False]
Length: 6, dtype: boolean

In [64]: df1[mask]
Out[64]:
          A         B         C         D
a  0.132003 -0.827317 -0.076467 -1.187678
c  1.024180  0.569605  0.875906 -2.211372
```

For getting a value explicitly:

```
# this is also equivalent to ``df1.at['a','A']``
In [65]: df1.loc['a', 'A']
Out[65]: np.float64(0.13200317033032932)
```

Slicing with labels

When using `.loc` with slices, if both the start and the stop labels are present in the index, then elements *located* between the two (including them) are returned:

```
In [66]: s = pd.Series(list('abcde'), index=[0, 3, 2, 5, 4])

In [67]: s.loc[3:5]
Out[67]:
3    b
2    c
5    d
dtype: str
```

If the index is sorted, and can be compared against start and stop labels, then slicing will still work as expected, by selecting labels which *rank* between the two:

```
In [68]: s.sort_index()
Out[68]:
0    a
2    c
3    b
4    e
5    d
dtype: str

In [69]: s.sort_index().loc[1:6]
Out[69]:
2    c
3    b
4    e
5    d
dtype: str
```

However, if at least one of the two is absent *and* the index is not sorted, an error will be raised (since doing otherwise would be computationally expensive, as well as potentially ambiguous for mixed type indexes). For instance, in the above example, `s.loc[1:6]` would raise `KeyError`.

For the rationale behind this behavior, see Endpoints are inclusive.

```
In [70]: s = pd.Series(list('abcdef'), index=[0, 3, 2, 5, 4, 2])

In [71]: s.loc[3:5]
Out[71]:
3    b
2    c
5    d
dtype: str
```

Also, if the index has duplicate labels *and* either the start or the stop label is duplicated, an error will be raised. For instance, in the above example, `s.loc[2:5]` would raise a `KeyError`.

For more information about duplicate labels, see Duplicate Labels.

When using a slice with a step, such as `.loc[start:stop:step]`, note that *start* and *stop* are interpreted as **labels**, while *step* is applied over the **positional index** within that label range. This means a stepped slice will behave differently than using the labels `range(start, stop, step)` when the index is not contiguous integers.

For example, in a `Series` with a non-contiguous integer index:

```
In [72]: s = pd.Series(range(10), index=[0, 5, 10, 15, 20, 25, 30, 35, 40, 45])

In [73]: s.loc[10:50:5]              # (10), then skip 3 positions → 35 only
Out[73]:
10    2
35    7
dtype: int64

In [74]: s.loc[[10, 15, 20, 25]]     # explicit label selection
Out[74]:
10    2
15    3
20    4
25    5
dtype: int64
```

The first applies *step* across **positional locations** between the start/stop labels. The second selects each label directly.

Similarly, with a string-based index, the behavior is identical:

```
In [75]: s = pd.Series(range(10), index=['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i

In [76]: s.loc['b':'i':2]              # Start at 'b' (position 1), stop at 'i' (posi
Out[76]:
b    1
d    3
f    5
h    7
dtype: int64

In [77]: s.loc[['b', 'd', 'f', 'h']] # explicit label selection
Out[77]:
b    1
d    3
f    5
h    7
dtype: int64
```

In both cases, *start* and *stop* determine the label boundaries (inclusive), while *step* skips positions within that range, regardless of the index type.

Selection by position

pandas provides a suite of methods in order to get **purely integer based indexing**. The semantics follow closely Python and NumPy slicing. These are `0-based` indexing. When slicing, the start bound is *included*, while the upper bound is *excluded*. Trying to use a non-integer, even a **valid** label will raise an `IndexError`.

The `.iloc` attribute is the primary access method. The following are valid inputs:

- An integer e.g. `5`.
- A list or array of integers `[4, 3, 0]`.
- A slice object with ints `1:7`.
- A boolean array.
- A `callable`, see Selection By Callable.
- A tuple of row (and column) indexes, whose elements are one of the above types.

```
In [78]: s1 = pd.Series(np.random.randn(5), index=list(range(0, 10, 2)))

In [79]: s1
Out[79]:
0    0.695775
2    0.341734
4    0.959726
6   -1.110336
8   -0.619976
dtype: float64

In [80]: s1.iloc[:3]
Out[80]:
0    0.695775
2    0.341734
4    0.959726
dtype: float64

In [81]: s1.iloc[3]
Out[81]: np.float64(-1.110336102891167)
```

Note that setting works as well:

```
In [82]: s1.iloc[:3] = 0

In [83]: s1
Out[83]:
0    0.000000
2    0.000000
4    0.000000
6   -1.110336
8   -0.619976
dtype: float64
```

With a DataFrame:

```
In [84]: df1 = pd.DataFrame(np.random.randn(6, 4),
   ....:                    index=list(range(0, 12, 2)),
   ....:                    columns=list(range(0, 8, 2)))
   ....: 

In [85]: df1
Out[85]:
           0         2         4         6
0   0.149748 -0.732339  0.687738  0.176444
2   0.403310 -0.154951  0.301624 -2.179861
4  -1.369849 -0.954208  1.462696 -1.743161
6  -0.826591 -0.345352  1.314232  0.690579
8   0.995761  2.396780  0.014871  3.357427
10 -0.317441 -1.236269  0.896171 -0.487602
```

Select via integer slicing:

```
In [86]: df1.iloc[:3]
Out[86]:
          0         2         4         6
0  0.149748 -0.732339  0.687738  0.176444
2  0.403310 -0.154951  0.301624 -2.179861
4 -1.369849 -0.954208  1.462696 -1.743161

In [87]: df1.iloc[1:5, 2:4]
Out[87]:
          4         6
2  0.301624 -2.179861
4  1.462696 -1.743161
6  1.314232  0.690579
8  0.014871  3.357427
```

Select via integer list:

```
In [88]: df1.iloc[[1, 3, 5], [1, 3]]
Out[88]:
           2         6
2  -0.154951 -2.179861
6  -0.345352  0.690579
10 -1.236269 -0.487602
```

```
In [89]: df1.iloc[1:3, :]
Out[89]:
          0         2         4         6
2  0.403310 -0.154951  0.301624 -2.179861
4 -1.369849 -0.954208  1.462696 -1.743161
```

```
In [90]: df1.iloc[:, 1:3]
Out[90]:
           2         4
0  -0.732339  0.687738
2  -0.154951  0.301624
4  -0.954208  1.462696
6  -0.345352  1.314232
8   2.396780  0.014871
10 -1.236269  0.896171
```

```
# this is also equivalent to ``df1.iat[1,1]``
In [91]: df1.iloc[1, 1]
Out[91]: np.float64(-0.1549507744249032)
```

For getting a cross section using an integer position (equiv to `df.xs(1)`):

```
In [92]: df1.iloc[1]
Out[92]:
0    0.403310
2   -0.154951
4    0.301624
6   -2.179861
Name: 2, dtype: float64
```

Out of range slice indexes are handled gracefully just as in Python/NumPy.

```
# these are allowed in Python/NumPy.
In [93]: x = list('abcdef')

In [94]: x
Out[94]: ['a', 'b', 'c', 'd', 'e', 'f']

In [95]: x[4:10]
Out[95]: ['e', 'f']

In [96]: x[8:10]
Out[96]: []

In [97]: s = pd.Series(x)

In [98]: s
Out[98]:
0    a
1    b
2    c
3    d
4    e
5    f
dtype: str

In [99]: s.iloc[4:10]
Out[99]:
4    e
5    f
dtype: str

In [100]: s.iloc[8:10]
Out[100]: Series([], dtype: str)
```

Note that using slices that go out of bounds can result in an empty axis (e.g. an empty DataFrame being returned).

```
In [101]: dfl = pd.DataFrame(np.random.randn(5, 2), columns=list('AB'))

In [102]: dfl
Out[102]:
          A         B
0 -0.082240 -2.182937
1  0.380396  0.084844
2  0.432390  1.519970
3 -0.493662  0.600178
4  0.274230  0.132885

In [103]: dfl.iloc[:, 2:3]
Out[103]:
Empty DataFrame
Columns: []
Index: [0, 1, 2, 3, 4]

In [104]: dfl.iloc[:, 1:3]
Out[104]:
          B
0 -2.182937
1  0.084844
2  1.519970
3  0.600178
4  0.132885

In [105]: dfl.iloc[4:6]
Out[105]:
         A         B
4  0.27423  0.132885
```

A single indexer that is out of bounds will raise an `IndexError`. A list of indexers where any element is out of bounds will raise an `IndexError`.

```
In [106]: dfl.iloc[[4, 5, 6]]
---------------------------------------------------------------------------
IndexError                                Traceback (most recent call last)
File ~/work/pandas/pandas/pandas/core/indexing.py:1735, in _iLocIndexer._get_list_a
   1734 try:
-> 1735     return self.obj.take(key, axis=axis)
   1736 except IndexError as err:
   1737     # re-raise with different error message, e.g. test_getitem_ndarray_3d

File ~/work/pandas/pandas/pandas/core/generic.py:4089, in NDFrame.take(self, indice
   4087         return self.copy(deep=False)
   4088 
-> 4089 new_data = self._mgr.take(
   4090     indices,

File ~/work/pandas/pandas/pandas/core/internals/managers.py:1087, in BaseBlockManag
   1086 n = self.shape[axis]
-> 1087 indexer = maybe_convert_indices(indexer, n, verify=verify)
   1089 new_labels = self.axes[axis].take(indexer)

File ~/work/pandas/pandas/pandas/core/indexers/utils.py:284, in maybe_convert_indic
    283     if mask.any():
--> 284         raise IndexError("indices are out-of-bounds")
    285 return indices

IndexError: indices are out-of-bounds

The above exception was the direct cause of the following exception:

IndexError                                Traceback (most recent call last)
Cell In[106], line 1
----> 1 dfl.iloc[[4, 5, 6]]

File ~/work/pandas/pandas/pandas/core/indexing.py:1207, in _LocationIndexer.__getite
   1205 maybe_callable = com.apply_if_callable(key, self.obj)
   1206 maybe_callable = self._raise_callable_usage(key, maybe_callable)
-> 1207 return self._getitem_axis(maybe_callable, axis=axis)

File ~/work/pandas/pandas/pandas/core/indexing.py:1764, in _iLocIndexer._getitem_ax
   1762 # a list of integers
   1763 elif is_list_like_indexer(key):
-> 1764     return self._get_list_axis(key, axis=axis)
   1766 # a single integer
   1767 else:
   1768     key = item_from_zerodim(key)

File ~/work/pandas/pandas/pandas/core/indexing.py:1738, in _iLocIndexer._get_list_a
   1735     return self.obj.take(key, axis=axis)
   1736 except IndexError as err:
   1737     # re-raise with different error message, e.g. test_getitem_ndarray_3d
-> 1738     raise IndexError("positional indexers are out-of-bounds") from err

IndexError: positional indexers are out-of-bounds
```

```
In [107]: dfl.iloc[:, 4]
---------------------------------------------------------------------------
IndexError                                Traceback (most recent call last)
Cell In[107], line 1
----> 1 dfl.iloc[:, 4]

File ~/work/pandas/pandas/pandas/core/indexing.py:1200, in _LocationIndexer.__getite
   1198     if self._is_scalar_access(key):
   1199         return self.obj._get_value(*key, takeable=self._takeable)
-> 1200     return self._getitem_tuple(key)
   1201 else:
   1202     # we by definition only have the 0th axis
   1203     axis = self.axis or 0

File ~/work/pandas/pandas/pandas/core/indexing.py:1711, in _iLocIndexer._getitem_tup
   1710 def _getitem_tuple(self, tup: tuple):
-> 1711     tup = self._validate_tuple_indexer(tup)
   1712     with suppress(IndexingError):
   1713         return self._getitem_lowerdim(tup)

File ~/work/pandas/pandas/pandas/core/indexing.py:993, in _LocationIndexer._validate
    991 for i, k in enumerate(key):
    992     try:
--> 993         self._validate_key(k, i)
    994     except ValueError as err:
    995         raise ValueError(
    996             "Location based indexing can only have "
    996             f"[{self._valid_types}] t
    997         ) from err

File ~/work/pandas/pandas/pandas/core/indexing.py:1605, in _iLocIndexer._validate_ke
   1603     return
   1604 elif is_integer(key):
-> 1605     self._validate_integer(key, axis)
   1606 elif isinstance(key, tuple):
   1607     # a tuple should already have been caught by this point
   1608     # so don't treat a tuple as a valid indexer
   1609     raise IndexingError("Too many indexers")

File ~/work/pandas/pandas/pandas/core/indexing.py:1706, in _iLocIndexer._validate_in
   1704 len_axis = len(self.obj._get_axis(axis))
   1705 if key >= len_axis or key < -len_axis:
-> 1706     raise IndexError("single positional indexer is out-of-bounds")

IndexError: single positional indexer is out-of-bounds
```

Selection by callable

`.loc`, `.iloc`, and also `[]` indexing can accept a `callable` as indexer. The `callable` must be a function with one argument (the calling Series or DataFrame) that returns valid output for indexing.

> **Note**
>
> For `.iloc` indexing, returning a tuple from the callable is not supported, since tuple destructuring for row and column indexes occurs *before* applying callables.

```
In [108]: df1 = pd.DataFrame(np.random.randn(6, 4),
   .....:                    index=list('abcdef'),
   .....:                    columns=list('ABCD'))
   .....: 

In [109]: df1
Out[109]:
          A         B         C         D
a -0.023688  2.410179  1.450520  0.206053
b -0.251905 -2.213588  1.063327  1.266143
c  0.299368 -0.863838  0.408204 -1.048089
d -0.025747 -0.988387  0.094055  1.262731
e  1.289997  0.082423 -0.055758  0.536580
f -0.489682  0.369374 -0.034571 -2.484478

In [110]: df1.loc[lambda df: df['A'] > 0, :]
Out[110]:
          A         B         C         D
c  0.299368 -0.863838  0.408204 -1.048089
e  1.289997  0.082423 -0.055758  0.536580

In [111]: df1.loc[:, lambda df: ['A', 'B']]
Out[111]:
          A         B
a -0.023688  2.410179
b -0.251905 -2.213588
c  0.299368 -0.863838
d -0.025747 -0.988387
e  1.289997  0.082423
f -0.489682  0.369374

In [112]: df1.iloc[:, lambda df: [0, 1]]
Out[112]:
          A         B
a -0.023688  2.410179
b -0.251905 -2.213588
c  0.299368 -0.863838
d -0.025747 -0.988387
e  1.289997  0.082423
f -0.489682  0.369374

In [113]: df1[lambda df: df.columns[0]]
Out[113]:
a   -0.023688
b   -0.251905
c    0.299368
d   -0.025747
e    1.289997
f   -0.489682
Name: A, dtype: float64
```

You can use callable indexing in `Series`.

```
In [114]: df1['A'].loc[lambda s: s > 0]
Out[114]:
c    0.299368
e    1.289997
Name: A, dtype: float64
```

Using these methods / indexers, you can chain data selection operations without using a temporary variable.

```
In [115]: bb = pd.read_csv('data/baseball.csv', index_col='id')

In [116]: (bb.groupby(['year', 'team']).sum(numeric_only=True)
   .....:    .loc[lambda df: df['r'] > 100])
   .....: 
Out[116]:
           stint    g    ab    r    h  X2b  ...     so   ibb   hbp    sh    sf  gid
year team                                    ...                                   
2007 CIN       6  379   745  101  203   35  ...  127.0  14.0   1.0   1.0  15.0  18.
     DET       5  301  1062  162  283   54  ...  176.0   3.0  10.0   4.0   8.0  28.
     HOU       4  311   926  109  218   47  ...  212.0   3.0   9.0  16.0   6.0  17.
     LAN      11  413  1021  153  293   61  ...  141.0   8.0   9.0   3.0   8.0  29.
     NYN      13  622  1854  240  509  101  ...  310.0  24.0  23.0  18.0  15.0  48.
     SFN       5  482  1305  198  337   67  ...  188.0  51.0   8.0  16.0   6.0  41.
     TEX       2  198   729  115  200   40  ...  140.0   4.0   5.0   2.0   8.0  16.
     TOR       4  459  1408  187  378   96  ...  265.0  16.0  12.0   4.0  16.0  38.

[8 rows x 18 columns]
```

Combining positional and label-based indexing

If you wish to get the 0th and the 2nd elements from the index in the 'A' column, you can do:

```
In [117]: dfd = pd.DataFrame({'A': [1, 2, 3],
   .....:                     'B': [4, 5, 6]},
   .....:                    index=list('abc'))
   .....: 

In [118]: dfd
Out[118]:
   A  B
a  1  4
b  2  5
c  3  6

In [119]: dfd.loc[dfd.index[[0, 2]], 'A']
Out[119]:
a    1
c    3
Name: A, dtype: int64
```

This can also be expressed using `.iloc`, by explicitly getting locations on the indexers, and using *positional* indexing to select things.

```
In [120]: dfd.iloc[[0, 2], dfd.columns.get_loc('A')]
Out[120]:
a    1
c    3
Name: A, dtype: int64
```

For getting *multiple* indexers, using `.get_indexer`:

```
In [121]: dfd.iloc[[0, 2], dfd.columns.get_indexer(['A', 'B'])]
Out[121]:
   A  B
a  1  4
c  3  6
```

Reindexing

The idiomatic way to achieve selecting potentially not-found elements is via `.reindex()`. See also the section on reindexing.

```
In [122]: s = pd.Series([1, 2, 3])

In [123]: s.reindex([1, 2, 3])
Out[123]:
1    2.0
2    3.0
3    NaN
dtype: float64
```

Alternatively, if you want to select only *valid* keys, the following is idiomatic and efficient; it is guaranteed to preserve the dtype of the selection.

```
In [124]: labels = [1, 2, 3]

In [125]: s.loc[s.index.intersection(labels)]
Out[125]:
1    2
2    3
dtype: int64
```

Having a duplicated index will raise for a `.reindex()`:

```
In [126]: s = pd.Series(np.arange(4), index=['a', 'a', 'b', 'c'])

In [127]: labels = ['c', 'd']

In [128]: s.reindex(labels)
---------------------------------------------------------------------------
ValueError                                Traceback (most recent call last)
Cell In[128], line 1
----> 1 s.reindex(labels)

File ~/work/pandas/pandas/pandas/core/series.py:5525, in Series.reindex(self, index
   5299 def reindex(  # type: ignore[override]
   5300     self,
   5301     index=None,
  (...)   5309     tolerance=None,
   5310 ) -> Series:
   5311     """
   5312     Conform Series to new index with optional filling logic.
   5313 
  (...)   5523         See the :ref:`user guide <basics.reindexing>` for more.
   5524     """
-> 5525     return super().reindex(
   5526         index=index,
   5527         method=method,
   5528         level=level,
   5529         fill_value=fill_value,
   5530         limit=limit,
   5531         tolerance=tolerance,
   5532         copy=copy,
   5533     )

File ~/work/pandas/pandas/pandas/core/generic.py:5464, in NDFrame.reindex(self, labe
   5460     if self._needs_reindex_multi(axes, method, level):
   5461         return self._reindex_multi(axes, fill_value)
   5462 
   5463     # perform the reindex on the axes
-> 5464     return self._reindex_axes(
   5465         axes, level, limit, tolerance, method, fill_value
   5466     ).__finalize__(self, method="reindex")

File ~/work/pandas/pandas/pandas/core/generic.py:5486, in NDFrame._reindex_axes(sel
   5482     if labels is None:
   5483         continue
   5484 
   5485     ax = self._get_axis(a)
-> 5486     new_index, indexer = ax.reindex(
   5487         labels, level=level, limit=limit, tolerance=tolerance, meth
   5488     )
   5489 

File ~/work/pandas/pandas/pandas/core/indexes/base.py:4253, in Index.reindex(self, 
   4250     raise ValueError("cannot handle a non-unique multi-index!")
   4251 elif not self.is_unique:
   4252     # GH#42568
-> 4253     raise ValueError("cannot reindex on an axis with duplicate labels")
   4254 else:
   4255     indexer, _ = self.get_indexer_non_unique(target)

ValueError: cannot reindex on an axis with duplicate labels
```

Generally, you can intersect the desired labels with the current axis, and then reindex.

```
In [129]: s.loc[s.index.intersection(labels)].reindex(labels)
Out[129]:
c    3.0
d    NaN
dtype: float64
```

However, this would *still* raise if your resulting index is duplicated.

```
In [130]: labels = ['a', 'd']

In [131]: s.loc[s.index.intersection(labels)].reindex(labels)
---------------------------------------------------------------------------
ValueError                                Traceback (most recent call last)
Cell In[131], line 1
----> 1 s.loc[s.index.intersection(labels)].reindex(labels)

File ~/work/pandas/pandas/pandas/core/series.py:5525, in Series.reindex(self, index
   5299 def reindex(  # type: ignore[override]
   5300     self,
   5301     index=None,
  (...)   5309     tolerance=None,
   5310 ) -> Series:
   5311     """
   5312     Conform Series to new index with optional filling logic.
   5313 
  (...)   5523         See the :ref:`user guide <basics.reindexing>` for more.
   5524     """
-> 5525     return super().reindex(
   5526         index=index,
   5527         method=method,
   5528         level=level,
   5529         fill_value=fill_value,
   5530         limit=limit,
   5531         tolerance=tolerance,
   5532         copy=copy,
   5533     )

File ~/work/pandas/pandas/pandas/core/generic.py:5464, in NDFrame.reindex(self, labe
   5460     if self._needs_reindex_multi(axes, method, level):
   5461         return self._reindex_multi(axes, fill_value)
   5462 
   5463     # perform the reindex on the axes
-> 5464     return self._reindex_axes(
   5465         axes, level, limit, tolerance, method, fill_value
   5466     ).__finalize__(self, method="reindex")

File ~/work/pandas/pandas/pandas/core/generic.py:5486, in NDFrame._reindex_axes(sel
   5482     if labels is None:
   5483         continue
   5484 
   5485     ax = self._get_axis(a)
-> 5486     new_index, indexer = ax.reindex(
   5487         labels, level=level, limit=limit, tolerance=tolerance, meth
   5488     )
   5489 

File ~/work/pandas/pandas/pandas/core/indexes/base.py:4253, in Index.reindex(self, 
   4250     raise ValueError("cannot handle a non-unique multi-index!")
   4251 elif not self.is_unique:
   4252     # GH#42568
-> 4253     raise ValueError("cannot reindex on an axis with duplicate labels")
   4254 else:
   4255     indexer, _ = self.get_indexer_non_unique(target)

ValueError: cannot reindex on an axis with duplicate labels
```

Selecting random samples

A random selection of rows or columns from a Series or DataFrame with the `sample()` method. The method will sample rows by default, and accepts a specific number of rows/columns to return, or a fraction of rows.

```
In [132]: s = pd.Series([0, 1, 2, 3, 4, 5])

# When no arguments are passed, returns 1 row.
In [133]: s.sample()
Out[133]:
4    4
dtype: int64

# One may specify either a number of rows:
In [134]: s.sample(n=3)
Out[134]:
0    0
4    4
1    1
dtype: int64

# Or a fraction of the rows:
In [135]: s.sample(frac=0.5)
Out[135]:
5    5
3    3
1    1
dtype: int64
```

By default, `sample` will return each row at most once, but one can also sample with replacement using the `replace` option:

```
In [136]: s = pd.Series([0, 1, 2, 3, 4, 5])

# Without replacement (default):
In [137]: s.sample(n=6, replace=False)
Out[137]:
0    0
1    1
5    5
3    3
2    2
4    4
dtype: int64

# With replacement:
In [138]: s.sample(n=6, replace=True)
Out[138]:
0    0
4    4
3    3
2    2
4    4
4    4
dtype: int64
```

By default, each row has an equal probability of being selected, but if you want rows to have different probabilities, you can pass the `sample` function sampling weights as `weights`. These weights can be a list, a NumPy array, or a Series, but they must be of the same length as the object you are sampling. Missing values will be treated as a weight of zero, and inf values are not allowed. If weights do not sum to 1, they will be re-normalized by dividing all weights by the sum of the weights. For example:

```
In [139]: s = pd.Series([0, 1, 2, 3, 4, 5])

In [140]: example_weights = [0, 0, 0.2, 0.2, 0.2, 0.4]

In [141]: s.sample(n=2, weights=example_weights)
Out[141]:
5    5
4    4
dtype: int64

# Weights will be re-normalized automatically
In [142]: example_weights2 = [0.5, 0, 0, 0, 0, 0]

In [143]: s.sample(n=1, weights=example_weights2)
Out[143]:
0    0
dtype: int64
```

When applied to a DataFrame, you can use a column of the DataFrame as sampling weights (provided you are sampling rows and not columns) by simply passing the name of the column as a string.

```
In [144]: df2 = pd.DataFrame({'col1': [9, 8, 7, 6],
   .....:                     'weight_column': [0.5, 0.4, 0.1, 0]})
   .....:

In [145]: df2.sample(n=2, weights='weight_column')
Out[145]:
   col1  weight_column
0     9            0.5
1     8            0.4
```

`sample` also allows users to sample columns instead of rows using the `axis` argument.

```
In [146]: df3 = pd.DataFrame({'col1': [1, 2, 3], 'col2': [2, 3, 4]})

In [147]: df3.sample(n=1, axis=1)
Out[147]:
   col2
0     2
1     3
2     4
```

Finally, one can also set a seed for `sample`'s random number generator using the `random_state` argument, which will accept either an integer (as a seed) or a NumPy RandomState object.

```
In [148]: df4 = pd.DataFrame({'col1': [1, 2, 3], 'col2': [2, 3, 4]})

# With a given seed, the sample will always draw the same rows.
In [149]: df4.sample(n=2, random_state=2)
Out[149]:
   col1  col2
2     3     4
1     2     3

In [150]: df4.sample(n=2, random_state=2)
Out[150]:
   col1  col2
2     3     4
1     2     3
```

Setting with enlargement

The `.loc/[]` operations can perform enlargement when setting a non-existent key for that axis.

In the `Series` case this is effectively an appending operation.

```
In [151]: se = pd.Series([1, 2, 3])

In [152]: se
Out[152]:
0    1
1    2
2    3
dtype: int64

In [153]: se[5] = 5.

In [154]: se
Out[154]:
0    1.0
1    2.0
2    3.0
5    5.0
dtype: float64
```

A `DataFrame` can be enlarged on either axis via `.loc`.

```
In [155]: dfi = pd.DataFrame(np.arange(6).reshape(3, 2),
   .....:                    columns=['A', 'B'])
   .....:

In [156]: dfi
Out[156]:
   A  B
0  0  1
1  2  3
2  4  5

In [157]: dfi.loc[:, 'C'] = dfi.loc[:, 'A']

In [158]: dfi
Out[158]:
   A  B  C
0  0  1  0
1  2  3  2
2  4  5  4
```

This is like an `append` operation on the `DataFrame`.

```
In [159]: dfi.loc[3] = 5

In [160]: dfi
Out[160]:
   A  B  C
0  0  1  0
1  2  3  2
2  4  5  4
3  5  5  5
```

Fast scalar value getting and setting

Since indexing with `[]` must handle a lot of cases (single-label access, slicing, boolean indexing, etc.), it has a bit of overhead in order to figure out what you're asking for. If you only want to access a scalar value, the fastest way is to use the `at` and `iat` methods, which are implemented on all of the data structures.

Similarly to `loc`, `at` provides **label** based scalar lookups, while, `iat` provides **integer** based lookups analogously to `iloc`

```
In [161]: s.iat[5]
Out[161]: np.int64(5)

In [162]: df.at[dates[5], 'A']
Out[162]: np.float64(0.1136484096888855)

In [163]: df.iat[3, 0]
Out[163]: np.float64(-0.7067711336300845)
```

You can also set using these same indexers.

```
In [164]: df.at[dates[5], 'E'] = 7

In [165]: df.iat[3, 0] = 7
```

`at` may enlarge the object in-place as above if the indexer is missing.

```
In [166]: df.at[dates[-1] + pd.Timedelta('1 day'), 0] = 7

In [167]: df
Out[167]:
                   A         B         C         D    E    0
2000-01-01 -0.282863  0.469112 -1.509059 -1.135632  NaN  NaN
2000-01-02 -0.173215  1.212112  0.119209 -1.044236  NaN  NaN
2000-01-03 -2.104569 -0.861849 -0.494929  1.071804  NaN  NaN
2000-01-04  7.000000  0.721555 -1.039575  0.271860  NaN  NaN
2000-01-05  0.567020 -0.424972  0.276232 -1.087401  NaN  NaN
2000-01-06  0.113648 -0.673690 -1.478427  0.524988  7.0  NaN
2000-01-07  0.577046  0.404705 -1.715002 -1.039268  NaN  NaN
2000-01-08 -1.157892 -0.370647 -1.344312  0.844885  NaN  NaN
2000-01-09       NaN       NaN       NaN       NaN  NaN  7.0
```

Boolean indexing

Another common operation is the use of boolean vectors to filter the data. The operators are: `|` for `or`, `&` for `and`, and `~` for `not`. These **must** be grouped by using parentheses, since by default Python will evaluate an expression such as `df['A'] > 2 & df['B'] < 3` as `df['A'] > (2 & df['B']) < 3`, while the desired evaluation order is `(df['A'] > 2) & (df['B'] < 3)`.

Using a boolean vector to index a Series works exactly as in a NumPy ndarray:

```
In [168]: s = pd.Series(range(-3, 4))

In [169]: s
Out[169]:
0   -3
1   -2
2   -1
3    0
4    1
5    2
6    3
dtype: int64

In [170]: s[s > 0]
Out[170]:
4    1
5    2
6    3
dtype: int64

In [171]: s[(s < -1) | (s > 0.5)]
Out[171]:
0   -3
1   -2
4    1
5    2
6    3
dtype: int64

In [172]: s[~(s < 0)]
Out[172]:
3    0
4    1
5    2
6    3
dtype: int64
```

You may select rows from a DataFrame using a boolean vector the same length as the DataFrame's index (for example, something derived from one of the columns of the DataFrame):

```
In [173]: df[df['A'] > 0]
Out[173]:
                   A         B         C         D    E   0
2000-01-04  7.000000  0.721555 -1.039575  0.271860  NaN NaN
2000-01-05  0.567020 -0.424972  0.276232 -1.087401  NaN NaN
2000-01-06  0.113648 -0.673690 -1.478427  0.524988  7.0 NaN
2000-01-07  0.577046  0.404705 -1.715002 -1.039268  NaN NaN
```

List comprehensions and the `map` method of Series can also be used to produce more complex criteria:

```
In [174]: df2 = pd.DataFrame({'a': ['one', 'one', 'two', 'three', 'two', 'one', 'si
   .....:                     'b': ['x', 'y', 'y', 'x', 'y', 'x', 'x'],
   .....:                     'c': np.random.randn(7)})
   .....:

# only want 'two' or 'three'
In [175]: criterion = df2['a'].map(lambda x: x.startswith('t'))

In [176]: df2[criterion]
Out[176]:
       a  b         c
2    two  y  2.543083
3  three  x  0.831311
4    two  y -0.816973

# equivalent but slower
In [177]: df2[[x.startswith('t') for x in df2['a']]]
Out[177]:
       a  b         c
2    two  y  2.543083
3  three  x  0.831311
4    two  y -0.816973

# Multiple criteria
In [178]: df2[criterion & (df2['b'] == 'x')]
Out[178]:
       a  b         c
3  three  x  0.831311
```

With the choice methods Selection by Label, Selection by Position, and Advanced Indexing you may select along more than one axis using boolean vectors combined with other indexing expressions.

```
In [179]: df2.loc[criterion & (df2['b'] == 'x'), 'b':'c']
Out[179]:
   b         c
3  x  0.831311
```

⚠ **Warning**

While `loc` supports two kinds of boolean indexing, `iloc` only supports indexing with a boolean array. If the indexer is a boolean `Series`, an error will be raised. For instance, in the following example, `df.iloc[s.values, 1]` is ok. The boolean indexer is an array. But `df.iloc[s, 1]` would raise `ValueError`.

```
In [180]: df = pd.DataFrame([[1, 2], [3, 4], [5, 6]],
   .....:                   index=list('abc'),
   .....:                   columns=['A', 'B'])
   .....:

In [181]: s = (df['A'] > 2)

In [182]: s
Out[182]:
a    False
b     True
c     True
Name: A, dtype: bool

In [183]: df.loc[s, 'B']
Out[183]:
b    4
c    6
Name: B, dtype: int64

In [184]: df.iloc[s.values, 1]
Out[184]:
b    4
c    6
Name: B, dtype: int64
```

Indexing with isin

Consider the `isin()` method of `Series`, which returns a boolean vector that is true wherever the `Series` elements exist in the passed list. This allows you to select rows where one or more columns have values you want:

```
In [185]: s = pd.Series(np.arange(5), index=np.arange(5)[::-1], dtype='int64')

In [186]: s
Out[186]:
4    0
3    1
2    2
1    3
0    4
dtype: int64

In [187]: s.isin([2, 4, 6])
Out[187]:
4    False
3    False
2     True
1    False
0     True
dtype: bool

In [188]: s[s.isin([2, 4, 6])]
Out[188]:
2    2
0    4
dtype: int64
```

The same method is available for `Index` objects and is useful for the cases when you don't know which of the sought labels are in fact present:

```
In [189]: s[s.index.isin([2, 4, 6])]
Out[189]:
4    0
2    2
dtype: int64

# compare it to the following
In [190]: s.reindex([2, 4, 6])
Out[190]:
2    2.0
4    0.0
6    NaN
dtype: float64
```

In addition to that, `MultiIndex` allows selecting a separate level to use in the membership check:

```
In [191]: s_mi = pd.Series(np.arange(6),
   .....:                  index=pd.MultiIndex.from_product([[0, 1], ['a', 'b', 'c'
   .....:

In [192]: s_mi
Out[192]:
0  a    0
   b    1
   c    2
1  a    3
   b    4
   c    5
dtype: int64

In [193]: s_mi.iloc[s_mi.index.isin([(1, 'a'), (2, 'b'), (0, 'c')])]
Out[193]:
0  c    2
1  a    3
dtype: int64

In [194]: s_mi.iloc[s_mi.index.isin(['a', 'c', 'e'], level=1)]
Out[194]:
0  a    0
   c    2
1  a    3
   c    5
dtype: int64
```

DataFrame also has an `isin()` method. When calling `isin`, pass a set of values as either an array or dict. If values is an array, `isin` returns a DataFrame of booleans that is the same shape as the original DataFrame, with True wherever the element is in the sequence of values.

```
In [195]: df = pd.DataFrame({'vals': [1, 2, 3, 4], 'ids': ['a', 'b', 'f', 'n'],
   .....:                    'ids2': ['a', 'n', 'c', 'n']})
   .....:

In [196]: values = ['a', 'b', 1, 3]

In [197]: df.isin(values)
Out[197]:
    vals    ids   ids2
0   True   True   True
1  False   True  False
2   True  False  False
3  False  False  False
```

Oftentimes you'll want to match certain values with certain columns. Just make values a `dict` where the key is the column, and the value is a list of items you want to check for.

```
In [198]: values = {'ids': ['a', 'b'], 'vals': [1, 3]}

In [199]: df.isin(values)
Out[199]:
    vals    ids   ids2
0   True   True  False
1  False   True  False
2   True  False  False
3  False  False  False
```

To return the DataFrame of booleans where the values are *not* in the original DataFrame, use the `~` operator:

```
In [200]: values = {'ids': ['a', 'b'], 'vals': [1, 3]}

In [201]: ~df.isin(values)
Out[201]:
    vals    ids  ids2
0  False  False  True
1   True  False  True
2  False   True  True
3   True   True  True
```

Combine DataFrame's `isin` with the `any()` and `all()` methods to quickly select subsets of your data that meet a given criteria. To select a row where each column meets its own criterion:

```
In [202]: values = {'ids': ['a', 'b'], 'ids2': ['a', 'c'], 'vals': [1, 3]}

In [203]: row_mask = df.isin(values).all(axis=1)

In [204]: df[row_mask]
Out[204]:
   vals ids ids2
0     1   a    a
```

The `where()` Method and Masking

Selecting values from a Series with a boolean vector generally returns a subset of the data. To guarantee that selection output has the same shape as the original data, you can use the `where` method in `Series` and `DataFrame`.

To return only the selected rows:

```
In [205]: s[s > 0]
Out[205]:
3    1
2    2
1    3
0    4
dtype: int64
```

To return a Series of the same shape as the original:

```
In [206]: s.where(s > 0)
Out[206]:
4    NaN
3    1.0
2    2.0
1    3.0
0    4.0
dtype: float64
```

Selecting values from a DataFrame with a boolean criterion now also preserves input data shape. `where` is used under the hood as the implementation. The code below is equivalent to `df.where(df < 0)`.

```
In [207]: dates = pd.date_range('1/1/2000', periods=8)

In [208]: df = pd.DataFrame(np.random.randn(8, 4),
   .....:                   index=dates, columns=['A', 'B', 'C', 'D'])
   .....:

In [209]: df[df < 0]
Out[209]:
                   A         B         C         D
2000-01-01       NaN       NaN -0.250643 -1.350999
2000-01-02       NaN -0.026679       NaN       NaN
2000-01-03       NaN -1.112060       NaN -1.281223
2000-01-04       NaN       NaN -0.592066 -0.650567
2000-01-05 -0.374599       NaN -1.133167       NaN
2000-01-06 -1.254148       NaN       NaN       NaN
2000-01-07 -0.524443 -0.712053 -0.267772       NaN
2000-01-08 -0.076848       NaN -1.819296 -1.122503
```

In addition, `where` takes an optional `other` argument for replacement of values where the condition is False, in the returned copy.

```
In [210]: df.where(df < 0, -df)
Out[210]:
                   A         B         C         D
2000-01-01 -0.368085 -0.224661 -0.250643 -1.350999
2000-01-02 -0.142692 -0.026679 -1.345835 -0.938848
2000-01-03 -0.509624 -1.112060 -1.648517 -1.281223
2000-01-04 -0.553689 -0.359996 -0.592066 -0.650567
2000-01-05 -0.374599 -0.592071 -1.133167 -0.661259
2000-01-06 -1.254148 -0.627193 -0.411295 -1.282903
2000-01-07 -0.524443 -0.712053 -0.267772 -1.762567
2000-01-08 -0.076848 -2.431230 -1.819296 -1.122503
```

You may wish to set values based on some boolean criteria. This can be done intuitively like so:

```
In [211]: s2 = s.copy()

In [212]: s2[s2 < 0] = 0

In [213]: s2
Out[213]:
4    0
3    1
2    2
1    3
0    4
dtype: int64

In [214]: df2 = df.copy()

In [215]: df2[df2 < 0] = 0

In [216]: df2
Out[216]:
                   A         B         C         D
2000-01-01  0.368085  0.224661  0.000000  0.000000
2000-01-02  0.142692  0.000000  1.345835  0.938848
2000-01-03  0.509624  0.000000  1.648517  0.000000
2000-01-04  0.553689  0.359996  0.000000  0.000000
2000-01-05  0.000000  0.592071  0.000000  0.661259
2000-01-06  0.000000  0.627193  0.411295  1.282903
2000-01-07  0.000000  0.000000  0.000000  1.762567
2000-01-08  0.000000  2.431230  0.000000  0.000000
```

`where` returns a modified copy of the data.

ℹ **Note**

The signature for `DataFrame.where()` differs from `numpy.where()`. Roughly `df1.where(m, df2)` is equivalent to `np.where(m, df1, df2)`.

```
In [217]: df.where(df < 0, -df) == np.where(df < 0, df, -df)
Out[217]:
               A     B     C     D
2000-01-01  True  True  True  True
2000-01-02  True  True  True  True
2000-01-03  True  True  True  True
2000-01-04  True  True  True  True
2000-01-05  True  True  True  True
2000-01-06  True  True  True  True
2000-01-07  True  True  True  True
2000-01-08  True  True  True  True
```

Alignment

Furthermore, `where` aligns the input boolean condition (ndarray or DataFrame), such that partial selection with setting is possible. This is analogous to partial setting via `.loc` (but on the contents rather than the axis labels).

```
In [218]: df2 = df.copy()

In [219]: df2[df2[1:4] > 0] = 3

In [220]: df2
Out[220]:
                   A         B         C         D
2000-01-01  0.368085  0.224661 -0.250643 -1.350999
2000-01-02  3.000000 -0.026679  3.000000  3.000000
2000-01-03  3.000000 -1.112060  3.000000 -1.281223
2000-01-04  3.000000  3.000000 -0.592066 -0.650567
2000-01-05 -0.374599  0.592071 -1.133167  0.661259
2000-01-06 -1.254148  0.627193  0.411295  1.282903
2000-01-07 -0.524443 -0.712053 -0.267772  1.762567
2000-01-08 -0.076848  2.431230 -1.819296 -1.122503
```

Where can also accept `axis` and `level` parameters to align the input when performing the `where`.

```
In [221]: df2 = df.copy()

In [222]: df2.where(df2 > 0, df2['A'], axis='index')
Out[222]:
                   A         B         C         D
2000-01-01  0.368085  0.224661  0.368085  0.368085
2000-01-02  0.142692  0.142692  1.345835  0.938848
2000-01-03  0.509624  0.509624  1.648517  0.509624
2000-01-04  0.553689  0.359996  0.553689  0.553689
2000-01-05 -0.374599  0.592071 -0.374599  0.661259
2000-01-06 -1.254148  0.627193  0.411295  1.282903
2000-01-07 -0.524443 -0.524443 -0.524443  1.762567
2000-01-08 -0.076848  2.431230 -0.076848 -0.076848
```

This is equivalent to (but faster than) the following.

```
In [223]: df2 = df.copy()

In [224]: df.apply(lambda x, y: x.where(x > 0, y), y=df['A'])
Out[224]:
                   A         B         C         D
2000-01-01  0.368085  0.224661  0.368085  0.368085
2000-01-02  0.142692  0.142692  1.345835  0.938848
2000-01-03  0.509624  0.509624  1.648517  0.509624
2000-01-04  0.553689  0.359996  0.553689  0.553689
2000-01-05 -0.374599  0.592071 -0.374599  0.661259
2000-01-06 -1.254148  0.627193  0.411295  1.282903
2000-01-07 -0.524443 -0.524443 -0.524443  1.762567
2000-01-08 -0.076848  2.431230 -0.076848 -0.076848
```

`where` can accept a callable as condition and `other` arguments. The function must be with one argument (the calling Series or DataFrame) and that returns valid output as condition and `other` argument.

```
In [225]: df3 = pd.DataFrame({'A': [1, 2, 3],
   .....:                     'B': [4, 5, 6],
   .....:                     'C': [7, 8, 9]})
   .....:

In [226]: df3.where(lambda x: x > 4, lambda x: x + 10)
Out[226]:
    A   B  C
0  11  14  7
1  12   5  8
2  13   6  9
```

Mask

`mask()` is the inverse boolean operation of `where`.

```
In [227]: s.mask(s >= 0)
Out[227]:
4   NaN
3   NaN
2   NaN
1   NaN
0   NaN
dtype: float64

In [228]: df.mask(df >= 0)
Out[228]:
                   A         B         C         D
2000-01-01       NaN       NaN -0.250643 -1.350999
2000-01-02       NaN -0.026679       NaN       NaN
2000-01-03       NaN -1.112060       NaN -1.281223
2000-01-04       NaN       NaN -0.592066 -0.650567
2000-01-05 -0.374599       NaN -1.133167       NaN
2000-01-06 -1.254148       NaN       NaN       NaN
2000-01-07 -0.524443 -0.712053 -0.267772       NaN
2000-01-08 -0.076848       NaN -1.819296 -1.122503
```

Setting with enlargement conditionally using `numpy()`

An alternative to `where()` is to use `numpy.where()`. Combined with setting a new column, you can use it to enlarge a DataFrame where the values are determined conditionally.

Consider you have two choices to choose from in the following DataFrame. And you want to set a new column color to 'green' when the second column has 'Z'. You can do the following:

```
In [229]: df = pd.DataFrame({'col1': list('ABBC'), 'col2': list('ZZXY')})

In [230]: df['color'] = np.where(df['col2'] == 'Z', 'green', 'red')

In [231]: df
Out[231]:
  col1 col2  color
0    A    Z  green
1    B    Z  green
2    B    X    red
3    C    Y    red
```

If you have multiple conditions, you can use `numpy.select()` to achieve that. Say corresponding to three conditions there are three choice of colors, with a fourth color as a fallback, you can do the following.

```
In [232]: conditions = [
   .....:     (df['col2'] == 'Z') & (df['col1'] == 'A'),
   .....:     (df['col2'] == 'Z') & (df['col1'] == 'B'),
   .....:     (df['col1'] == 'B')
   .....: ]
   .....:
```

```
In [233]: choices = ['yellow', 'blue', 'purple']

In [234]: df['color'] = np.select(conditions, choices, default='black')

In [235]: df
Out[235]:
  col1 col2   color
0    A    Z  yellow
1    B    Z    blue
2    B    X  purple
3    C    Y   black
```

The `query()` Method

`DataFrame` objects have a `query()` method that allows selection using an expression.

You can get the value of the frame where column `b` has values between the values of columns `a` and `c`. For example:

```
In [236]: n = 10

In [237]: df = pd.DataFrame(np.random.rand(n, 3), columns=list('abc'))

In [238]: df
Out[238]:
          a         b         c
0  0.977227  0.727376  0.630865
1  0.076462  0.474453  0.438921
2  0.118680  0.863670  0.138138
3  0.577363  0.686602  0.595307
4  0.564592  0.520630  0.913052
5  0.926075  0.616184  0.078718
6  0.854477  0.898725  0.076404
7  0.523211  0.591538  0.792342
8  0.216974  0.564056  0.397890
9  0.454131  0.915716  0.074315

# pure python
In [239]: df[(df['a'] < df['b']) & (df['b'] < df['c'])]
Out[239]:
          a         b         c
7  0.523211  0.591538  0.792342

# query
In [240]: df.query('(a < b) & (b < c)')
Out[240]:
          a         b         c
7  0.523211  0.591538  0.792342
```

Do the same thing but fall back on a named index if there is no column with the name `a`.

```
In [241]: df = pd.DataFrame(np.random.randint(n / 2, size=(n, 2)), columns=list('bc

In [242]: df.index.name = 'a'

In [243]: df
Out[243]:
   b  c
a
0  0  0
1  3  1
2  3  4
3  0  4
4  0  1
5  3  4
6  4  3
7  1  4
8  0  3
9  0  1

In [244]: df.query('a < b and b < c')
Out[244]:
   b  c
a
2  3  4
```

If instead you don’t want to or cannot name your index, you can use the name `index` in your query expression:

```
In [245]: df = pd.DataFrame(np.random.randint(n, size=(n, 2)), columns=list('bc'))

In [246]: df
Out[246]:
   b  c
0  2  3
1  9  1
2  3  1
3  3  0
4  5  6
5  5  2
6  7  4
7  0  1
8  2  5
9  0  1

In [247]: df.query('index < b < c')
Out[247]:
   b  c
0  2  3
4  5  6
```

> **Note**
>
> If the name of your index overlaps with a column name, the column name is given precedence. For example,
>
> ```
> In [248]: df = pd.DataFrame({'a': np.random.randint(5, size=5)})
>
> In [249]: df.index.name = 'a'
>
> In [250]: df.query('a > 2')  # uses the column 'a', not the index
> Out[250]:
>    a
> a
> 3  3
> ```
>
> You can still use the index in a query expression by using the special identifier ‘index’:
>
> ```
> In [251]: df.query('index > 2')
> Out[251]:
>    a
> a
> 3  3
> 4  1
> ```
>
> If for some reason you have a column named `index`, then you can refer to the index as `ilevel_0` as well, but at this point you should consider renaming your columns to something less ambiguous.

`MultiIndex` `query()` Syntax

You can also use the levels of a `DataFrame` with a `MultiIndex` as if they were columns in the frame:

```
In [252]: n = 10

In [253]: colors = np.random.choice(['red', 'green'], size=n)

In [254]: foods = np.random.choice(['eggs', 'ham'], size=n)

In [255]: colors
Out[255]:
array(['green', 'green', 'red', 'green', 'red', 'red', 'red', 'red',
       'green', 'green'], dtype='<U5')

In [256]: foods
Out[256]:
array(['ham', 'ham', 'ham', 'ham', 'ham', 'ham', 'ham', 'eggs', 'eggs',
       'eggs'], dtype='<U4')

In [257]: index = pd.MultiIndex.from_arrays([colors, foods], names=['color', 'food'

In [258]: df = pd.DataFrame(np.random.randn(n, 2), index=index)

In [259]: df
Out[259]:
                   0         1
color food
green ham  -1.087664 -0.883833
      ham  -1.554827 -0.118953
red   ham  -1.460084 -0.020351
green ham  -0.256125  0.358575
red   ham   1.112033 -0.200521
      ham  -0.508784 -0.327758
      ham   0.627056  0.067058
      eggs -1.376511  1.162330
green eggs -0.482120 -0.455309
      eggs -0.985682  1.383438

In [260]: df.query('color == "red"')
Out[260]:
                   0         1
color food
red   ham  -1.460084 -0.020351
      ham   1.112033 -0.200521
      ham  -0.508784 -0.327758
      ham   0.627056  0.067058
      eggs -1.376511  1.162330
```

If the levels of the `MultiIndex` are unnamed, you can refer to them using special names:

```
In [261]: df.index.names = [None, None]

In [262]: df
Out[262]:
                   0         1
green ham  -1.087664 -0.883833
      ham  -1.554827 -0.118953
red   ham  -1.460084 -0.020351
green ham  -0.256125  0.358575
red   ham   1.112033 -0.200521
      ham  -0.508784 -0.327758
      ham   0.627056  0.067058
      eggs -1.376511  1.162330
green eggs -0.482120 -0.455309
      eggs -0.985682  1.383438

In [263]: df.query('ilevel_0 == "red"')
Out[263]:
                 0         1
red ham  -1.460084 -0.020351
    ham   1.112033 -0.200521
    ham  -0.508784 -0.327758
    ham   0.627056  0.067058
    eggs -1.376511  1.162330
```

The convention is `ilevel_0`, which means “index level 0” for the 0th level of the `index`.

`query()` Use Cases

A use case for `query()` is when you have a collection of `DataFrame` objects that have a subset of column names (or index levels/names) in common. You can pass the same query to both frames *without* having to specify which frame you’re interested in querying

```
In [264]: df = pd.DataFrame(np.random.rand(n, 3), columns=list('abc'))

In [265]: df
Out[265]:
          a         b         c
0  0.972314  0.789179  0.293847
1  0.374439  0.739133  0.221186
2  0.900625  0.534438  0.608763
3  0.166933  0.731582  0.965147
4  0.763981  0.372737  0.639792
5  0.702270  0.730804  0.134089
6  0.522758  0.311910  0.656542
7  0.258647  0.655096  0.654920
8  0.452594  0.454307  0.918260
9  0.581556  0.470410  0.417434

In [266]: df2 = pd.DataFrame(np.random.rand(n + 2, 3), columns=df.columns)

In [267]: df2
Out[267]:
           a         b         c
0   0.552021  0.483125  0.807046
1   0.277950  0.213500  0.471524
2   0.501458  0.141708  0.763617
3   0.081639  0.906284  0.480101
4   0.472250  0.380061  0.822149
5   0.459151  0.851196  0.125791
6   0.857816  0.795472  0.527728
7   0.561164  0.945324  0.622249
8   0.511283  0.577675  0.989138
9   0.528050  0.627750  0.652326
10  0.393289  0.103627  0.056786
11  0.749255  0.505533  0.883673

In [268]: expr = '0.0 <= a <= c <= 0.5'

In [269]: map(lambda frame: frame.query(expr), [df, df2])
Out[269]: <map at 0x7f5f64c65f60>
```

`query()` Python versus pandas Syntax Comparison

Full numpy-like syntax:

```
In [270]: df = pd.DataFrame(np.random.randint(n, size=(n, 3)), columns=list('abc'))

In [271]: df
Out[271]:
   a  b  c
0  6  2  2
1  2  6  3
2  3  8  2
3  1  7  2
4  5  1  5
5  9  8  0
6  1  5  0
7  4  9  6
8  2  3  0
9  6  5  4

In [272]: df.query('(a < b) & (b < c)')
Out[272]:
Empty DataFrame
Columns: [a, b, c]
Index: []

In [273]: df[(df['a'] < df['b']) & (df['b'] < df['c'])]
Out[273]:
Empty DataFrame
Columns: [a, b, c]
Index: []
```

Slightly nicer by removing the parentheses (comparison operators bind tighter than `&` and `|`):

```
In [274]: df.query('a < b & b < c')
Out[274]:
Empty DataFrame
Columns: [a, b, c]
Index: []
```

Use English instead of symbols:

```
In [275]: df.query('a < b and b < c')
Out[275]:
Empty DataFrame
Columns: [a, b, c]
Index: []
```

Pretty close to how you might write it on paper:

```
In [276]: df.query('a < b < c')
Out[276]:
Empty DataFrame
Columns: [a, b, c]
Index: []
```

The `in` and `not in` operators

`query()` also supports special use of Python’s `in` and `not in` comparison operators, providing a succinct syntax for calling the `isin` method of a `Series` or `DataFrame`.

```
# get all rows where columns "a" and "b" have overlapping values
In [277]: df = pd.DataFrame({'a': list('aabbccddeeff'), 'b': list('aaaabbbbcccc'),
   .....:                    'c': np.random.randint(5, size=12),
   .....:                    'd': np.random.randint(9, size=12)})
   .....:

In [278]: df
Out[278]:
    a  b  c  d
0   a  a  2  3
1   a  a  0  0
2   b  a  1  6
3   b  a  2  2
4   c  b  1  7
5   c  b  2  7
6   d  b  3  6
7   d  b  2  7
8   e  c  2  8
9   e  c  2  8
10  f  c  3  3
11  f  c  1  3

In [279]: df.query('a in b')
Out[279]:
   a  b  c  d
0  a  a  2  3
1  a  a  0  0
2  b  a  1  6
3  b  a  2  2
4  c  b  1  7
5  c  b  2  7

# How you'd do it in pure Python
In [280]: df[df['a'].isin(df['b'])]
Out[280]:
   a  b  c  d
0  a  a  2  3
1  a  a  0  0
2  b  a  1  6
3  b  a  2  2
4  c  b  1  7
5  c  b  2  7

In [281]: df.query('a not in b')
Out[281]:
    a  b  c  d
6   d  b  3  6
7   d  b  2  7
8   e  c  2  8
9   e  c  2  8
10  f  c  3  3
11  f  c  1  3

# pure Python
In [282]: df[~df['a'].isin(df['b'])]
Out[282]:
    a  b  c  d
6   d  b  3  6
7   d  b  2  7
8   e  c  2  8
9   e  c  2  8
10  f  c  3  3
11  f  c  1  3
```

You can combine this with other expressions for very succinct queries:

```
# rows where cols a and b have overlapping values
# and col c's values are less than col d's
In [283]: df.query('a in b and c < d')
Out[283]:
   a  b  c  d
0  a  a  2  3
2  b  a  1  6
4  c  b  1  7
5  c  b  2  7

# pure Python
In [284]: df[df['b'].isin(df['a']) & (df['c'] < df['d'])]
Out[284]:
    a  b  c  d
0   a  a  2  3
2   b  a  1  6
4   c  b  1  7
5   c  b  2  7
6   d  b  3  6
7   d  b  2  7
8   e  c  2  8
9   e  c  2  8
11  f  c  1  3
```

> **Note**
>
> Note that `in` and `not in` are evaluated in Python, since `numexpr` has no equivalent of this operation. However, **only the** `in`/`not in` **expression itself** is evaluated in vanilla Python. For example, in the expression
>
> ```
> df.query('a in b + c + d')
> ```
>
> `(b + c + d)` is evaluated by `numexpr` and *then* the `in` operation is evaluated in plain Python. In general, any operations that can be evaluated using `numexpr` will be.

Special use of the `==` operator with `list` objects

Comparing a `list` of values to a column using `==`/`!=` works similarly to `in`/`not in`.

```
In [285]: df.query('b == ["a", "b", "c"]')
Out[285]:
    a  b  c  d
0   a  a  2  3
1   a  a  0  0
2   b  a  1  6
3   b  a  2  2
4   c  b  1  7
5   c  b  2  7
6   d  b  3  6
7   d  b  2  7
8   e  c  2  8
9   e  c  2  8
10  f  c  3  3
11  f  c  1  3

# pure Python
In [286]: df[df['b'].isin(["a", "b", "c"])]
Out[286]:
    a  b  c  d
0   a  a  2  3
1   a  a  0  0
2   b  a  1  6
3   b  a  2  2
4   c  b  1  7
5   c  b  2  7
6   d  b  3  6
7   d  b  2  7
8   e  c  2  8
9   e  c  2  8
10  f  c  3  3
11  f  c  1  3

In [287]: df.query('c == [1, 2]')
Out[287]:
    a  b  c  d
0   a  a  2  3
2   b  a  1  6
3   b  a  2  2
4   c  b  1  7
5   c  b  2  7
7   d  b  2  7
8   e  c  2  8
9   e  c  2  8
11  f  c  1  3

In [288]: df.query('c != [1, 2]')
Out[288]:
    a  b  c  d
1   a  a  0  0
6   d  b  3  6
10  f  c  3  3

# using in/not in
In [289]: df.query('[1, 2] in c')
Out[289]:
    a  b  c  d
0   a  a  2  3
2   b  a  1  6
3   b  a  2  2
4   c  b  1  7
5   c  b  2  7
7   d  b  2  7
8   e  c  2  8
9   e  c  2  8
11  f  c  1  3

In [290]: df.query('[1, 2] not in c')
Out[290]:
    a  b  c  d
1   a  a  0  0
6   d  b  3  6
10  f  c  3  3

# pure Python
In [291]: df[df['c'].isin([1, 2])]
Out[291]:
    a  b  c  d
0   a  a  2  3
2   b  a  1  6
3   b  a  2  2
4   c  b  1  7
5   c  b  2  7
7   d  b  2  7
8   e  c  2  8
9   e  c  2  8
11  f  c  1  3
```

Boolean operators

You can negate boolean expressions with the word `not` or the `~` operator.

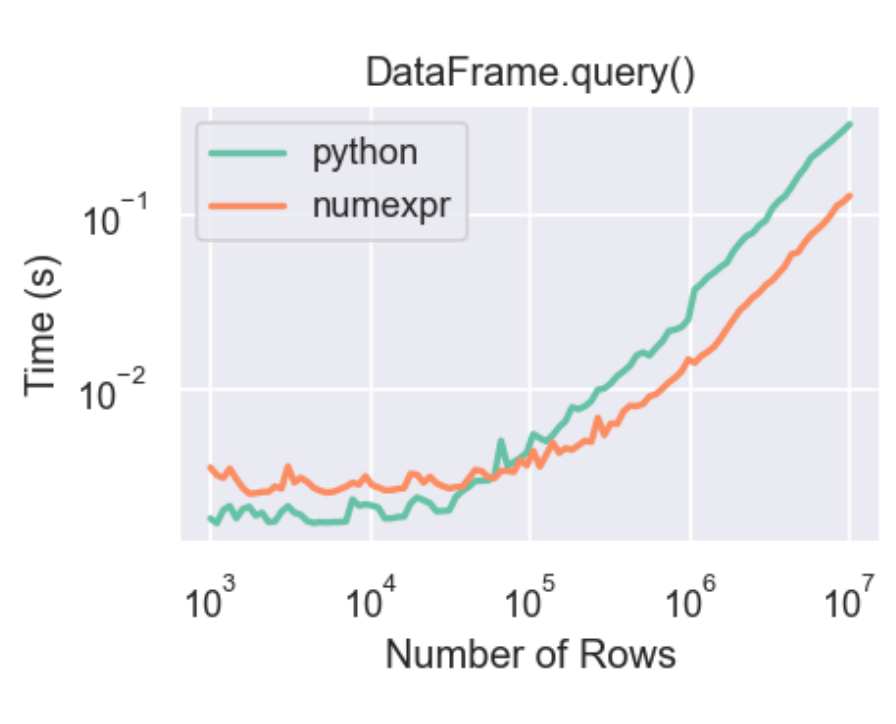
```
In [292]: df = pd.DataFrame(np.random.rand(n, 3), columns=list('abc'))

In [293]: df['bools'] = np.random.rand(len(df)) > 0.5

In [294]: df.query('~bools')
Out[294]:
          a         b         c  bools
4  0.580078  0.620439  0.434445  False
7  0.260776  0.290751  0.462080  False
9  0.165948  0.902260  0.825007  False

In [295]: df.query('not bools')
Out[295]:
          a         b         c  bools
4  0.580078  0.620439  0.434445  False
7  0.260776  0.290751  0.462080  False
9  0.165948  0.902260  0.825007  False

In [296]: df.query('not bools') == df[~df['bools']]
Out[296]:
      a     b     c  bools
4  True  True  True   True
7  True  True  True   True
9  True  True  True   True
```

Of course, expressions can be arbitrarily complex too:

```
# short query syntax
In [297]: shorter = df.query('a < b < c and (not bools) or bools > 2')

# equivalent in pure Python
In [298]: longer = df[(df['a'] < df['b'])
   .....:             & (df['b'] < df['c'])
   .....:             & (~df['bools'])
   .....:             | (df['bools'] > 2)]
   .....:

In [299]: shorter
Out[299]:
          a         b        c  bools
7  0.260776  0.290751  0.46208  False

In [300]: longer
Out[300]:
          a         b        c  bools
7  0.260776  0.290751  0.46208  False

In [301]: shorter == longer
Out[301]:
      a     b     c  bools
7  True  True  True   True
```

Performance of `query()`

`DataFrame.query()` using `numexpr` is slightly faster than Python for large frames.



You will only see the performance benefits of using the `numexpr` engine with `DataFrame.query()` if your frame has more than approximately 100,000 rows.

This plot was created using a `DataFrame` with 3 columns each containing floating point values generated using `numpy.random.randn()`.

```
In [302]: df = pd.DataFrame(np.random.randn(8, 4),
   .....:                   index=dates, columns=['A', 'B', 'C', 'D'])
   .....:

In [303]: df2 = df.copy()
```

Duplicate data

If you want to identify and remove duplicate rows in a DataFrame, there are two methods that will help: `duplicated` and `drop_duplicates`. Each takes as an argument the columns to use to identify duplicated rows.

- `duplicated` returns a boolean vector whose length is the number of rows, and which indicates whether a row is duplicated.
- `drop_duplicates` removes duplicate rows.

By default, the first observed row of a duplicate set is considered unique, but each method has a `keep` parameter to specify targets to be kept.

- `keep='first'` (default): mark / drop duplicates except for the first occurrence.
- `keep='last'`: mark / drop duplicates except for the last occurrence.
- `keep=False`: mark / drop all duplicates.

```
In [304]: df2 = pd.DataFrame({'a': ['one', 'one', 'two', 'two', 'two', 'three', 'fo
   .....:                     'b': ['x', 'y', 'x', 'y', 'x', 'x', 'x'],
   .....:                     'c': np.random.randn(7)})
   .....:

In [305]: df2
Out[305]:
       a  b         c
0    one  x -1.467157
1    one  y  0.113937
2    two  x -1.428572
3    two  y  0.337109
4    two  x  0.052469
5  three  x -2.294560
6   four  x  2.148507

In [306]: df2.duplicated('a')
Out[306]:
0    False
1     True
2    False
3     True
4     True
5    False
6    False
dtype: bool

In [307]: df2.duplicated('a', keep='last')
Out[307]:
0     True
1    False
2     True
3     True
4    False
5    False
6    False
dtype: bool

In [308]: df2.duplicated('a', keep=False)
Out[308]:
0     True
1     True
2     True
3     True
4     True
5    False
6    False
dtype: bool

In [309]: df2.drop_duplicates('a')
Out[309]:
       a  b         c
0    one  x -1.467157
2    two  x -1.428572
5  three  x -2.294560
6   four  x  2.148507

In [310]: df2.drop_duplicates('a', keep='last')
Out[310]:
       a  b         c
1    one  y  0.113937
4    two  x  0.052469
5  three  x -2.294560
6   four  x  2.148507

In [311]: df2.drop_duplicates('a', keep=False)
Out[311]:
       a  b         c
5  three  x -2.294560
6   four  x  2.148507
```

Also, you can pass a list of columns to identify duplications.

```
In [312]: df2.duplicated(['a', 'b'])
Out[312]:
0    False
1    False
2    False
3    False
4     True
5    False
6    False
dtype: bool

In [313]: df2.drop_duplicates(['a', 'b'])
Out[313]:
       a  b         c
0    one  x -1.467157
1    one  y  0.113937
2    two  x -1.428572
3    two  y  0.337109
5  three  x -2.294560
6   four  x  2.148507
```

To drop duplicates by index value, use `Index.duplicated` then perform slicing. The same set of options are available for the `keep` parameter.

```
In [314]: df3 = pd.DataFrame({'a': np.arange(6),
   .....:                     'b': np.random.randn(6)},
   .....:                    index=['a', 'a', 'b', 'c', 'b', 'a'])
   .....:

In [315]: df3
Out[315]:
   a         b
a  0 -0.293144
a  1 -1.596615
b  2  0.149716
c  3  0.173897
b  4 -0.049440
a  5  1.394590

In [316]: df3.index.duplicated()
Out[316]: array([False,  True, False, False,  True,  True])

In [317]: df3[~df3.index.duplicated()]
Out[317]:
   a         b
a  0 -0.293144
b  2  0.149716
c  3  0.173897

In [318]: df3[~df3.index.duplicated(keep='last')]
Out[318]:
   a         b
c  3  0.173897
b  4 -0.049440
a  5  1.394590
```

```
In [319]: df3[~df3.index.duplicated(keep=False)]
Out[319]: 
   a         b
c  3  0.173897
```

Dictionary-like `get()` method

Each of Series or DataFrame have a `get` method which can return a default value.

```
In [320]: s = pd.Series([1, 2, 3], index=['a', 'b', 'c'])

In [321]: s.get('a')  # equivalent to s['a']
Out[321]: np.int64(1)

In [322]: s.get('x', default=-1)
Out[322]: -1
```

Looking up values by index/column labels

Sometimes you want to extract a set of values given a sequence of row labels and column labels, this can be achieved by `pandas.factorize` and NumPy indexing.

For heterogeneous column types, we subset columns to avoid unnecessary NumPy conversions:

```
def pd_lookup_het(df, row_labels, col_labels):
    rows = df.index.get_indexer(row_labels)
    cols = df.columns.get_indexer(col_labels)
    sub = df.take(np.unique(cols), axis=1)
    sub = sub.take(np.unique(rows), axis=0)
    rows = sub.index.get_indexer(row_labels)
    values = sub.melt()["value"]
    cols = sub.columns.get_indexer(col_labels)
    flat_index = rows + cols * len(sub)
    result = values[flat_index]
    return result
```

For homogeneous column types, it is fastest to skip column subsetting and go directly to NumPy:

```
def pd_lookup_hom(df, row_labels, col_labels):
    rows = df.index.get_indexer(row_labels)
    df = df.loc[:, sorted(set(col_labels))]
    cols = df.columns.get_indexer(col_labels)
    result = df.to_numpy()[rows, cols]
    return result
```

Formerly this could be achieved with the dedicated `DataFrame.lookup` method which was deprecated in version 1.2.0 and removed in version 2.0.0.

Index objects

The pandas **`Index`** class and its subclasses can be viewed as implementing an *ordered multiset*. Duplicates are allowed.

`Index` also provides the infrastructure necessary for lookups, data alignment, and reindexing. The easiest way to create an **`Index`** directly is to pass a `list` or other sequence to **`Index`**:

```
In [323]: index = pd.Index(['e', 'd', 'a', 'b'])

In [324]: index
Out[324]: Index(['e', 'd', 'a', 'b'], dtype='str')

In [325]: 'd' in index
Out[325]: True
```

or using numbers:

```
In [326]: index = pd.Index([1, 5, 12])

In [327]: index
Out[327]: Index([1, 5, 12], dtype='int64')

In [328]: 5 in index
Out[328]: True
```

If no dtype is given, `Index` tries to infer the dtype from the data. It is also possible to give an explicit dtype when instantiating an **`Index`**:

```
In [329]: index = pd.Index(['e', 'd', 'a', 'b'], dtype="string")

In [330]: index
Out[330]: Index(['e', 'd', 'a', 'b'], dtype='string')

In [331]: index = pd.Index([1, 5, 12], dtype="int8")

In [332]: index
Out[332]: Index([1, 5, 12], dtype='int8')

In [333]: index = pd.Index([1, 5, 12], dtype="float32")

In [334]: index
Out[334]: Index([1.0, 5.0, 12.0], dtype='float32')
```

You can also pass a `name` to be stored in the index:

```
In [335]: index = pd.Index(['e', 'd', 'a', 'b'], name='something')

In [336]: index.name
Out[336]: 'something'
```

The name, if set, will be shown in the console display:

```
In [337]: index = pd.Index(list(range(5)), name='rows')

In [338]: columns = pd.Index(['A', 'B', 'C'], name='cols')

In [339]: df = pd.DataFrame(np.random.randn(5, 3), index=index, columns=columns)

In [340]: df
Out[340]: 
cols         A         B         C
rows                              
0     0.698035  0.631397  0.816412
1     0.709404 -1.193616 -0.263520
2    -0.878602  0.035458 -0.285808
3    -0.957431  2.243279 -1.124957
4    -1.994374  0.050270  0.512794

In [341]: df['A']
Out[341]: 
rows
0    0.698035
1    0.709404
2   -0.878602
3   -0.957431
4   -1.994374
Name: A, dtype: float64
```

Setting metadata

Indexes are "mostly immutable", but it is possible to set and change their `name` attribute. You can use the `rename`, `set_names` to set these attributes directly, and they default to returning a copy.

See Advanced Indexing for usage of MultiIndexes.

```
In [342]: ind = pd.Index([1, 2, 3])

In [343]: ind.rename("apple")
Out[343]: Index([1, 2, 3], dtype='int64', name='apple')

In [344]: ind
Out[344]: Index([1, 2, 3], dtype='int64')

In [345]: ind = ind.set_names(["apple"])

In [346]: ind.name = "bob"

In [347]: ind
Out[347]: Index([1, 2, 3], dtype='int64', name='bob')
```

`set_names`, `set_levels`, and `set_codes` also take an optional `level` argument

```
In [348]: index = pd.MultiIndex.from_product([range(3), ['one', 'two']], names=['fi

In [349]: index
Out[349]: 
MultiIndex([(0, 'one'),
            (0, 'two'),
            (1, 'one'),
            (1, 'two'),
            (2, 'one'),
            (2, 'two')],
           names=['first', 'second'])

In [350]: index.levels[1]
Out[350]: Index(['one', 'two'], dtype='str', name='second')

In [351]: index.set_levels(["a", "b"], level=1)
Out[351]: 
MultiIndex([(0, 'a'),
            (0, 'b'),
            (1, 'a'),
            (1, 'b'),
            (2, 'a'),
            (2, 'b')],
           names=['first', 'second'])
```

Set operations on Index objects

The two main operations are `union` and `intersection`. Difference is provided via the `.difference()` method.

```
In [352]: a = pd.Index(['c', 'b', 'a'])

In [353]: b = pd.Index(['c', 'e', 'd'])

In [354]: a.difference(b)
Out[354]: Index(['a', 'b'], dtype='str')
```

Also available is the `symmetric_difference` operation, which returns elements that appear in either `idx1` or `idx2`, but not in both. This is equivalent to the Index created by `idx1.difference(idx2).union(idx2.difference(idx1))`, with duplicates dropped.

```
In [355]: idx1 = pd.Index([1, 2, 3, 4])

In [356]: idx2 = pd.Index([2, 3, 4, 5])

In [357]: idx1.symmetric_difference(idx2)
Out[357]: Index([1, 5], dtype='int64')
```

> **Note**
>
> The resulting index from a set operation will be sorted in ascending order.

When performing **`Index.union()`** between indexes with different dtypes, the indexes must be cast to a common dtype. Typically, though not always, this is object dtype. The exception is when performing a union between integer and float data. In this case, the integer values are converted to float

```
In [358]: idx1 = pd.Index([0, 1, 2])

In [359]: idx2 = pd.Index([0.5, 1.5])

In [360]: idx1.union(idx2)
Out[360]: Index([0.0, 0.5, 1.0, 1.5, 2.0], dtype='float64')
```

Missing values

> **Important**
>
> Even though `Index` can hold missing values (`NaN`), it should be avoided if you do not want any unexpected results. For example, some operations exclude missing values implicitly.

`Index.fillna` fills missing values with specified scalar value.

```
In [361]: idx1 = pd.Index([1, np.nan, 3, 4])

In [362]: idx1
Out[362]: Index([1.0, nan, 3.0, 4.0], dtype='float64')

In [363]: idx1.fillna(2)
Out[363]: Index([1.0, 2.0, 3.0, 4.0], dtype='float64')

In [364]: idx2 = pd.DatetimeIndex([pd.Timestamp('2011-01-01'),
   .....:                          pd.NaT,
   .....:                          pd.Timestamp('2011-01-03')])
   .....: 

In [365]: idx2
Out[365]: DatetimeIndex(['2011-01-01', 'NaT', '2011-01-03'], dtype='datetime64[us]'

In [366]: idx2.fillna(pd.Timestamp('2011-01-02'))
Out[366]: DatetimeIndex(['2011-01-01', '2011-01-02', '2011-01-03'], dtype='datetime
```

Set / reset index

Occasionally you will load or create a data set into a DataFrame and want to add an index after you've already done so. There are a couple of different ways.

Set an index

DataFrame has a **`set_index()`** method which takes a column name (for a regular `Index`) or a list of column names (for a `MultiIndex`). To create a new, re-indexed DataFrame:

```
In [367]: data = pd.DataFrame({'a': ['bar', 'bar', 'foo', 'foo'],
   .....:                      'b': ['one', 'two', 'one', 'two'],
   .....:                      'c': ['z', 'y', 'x', 'w'],
   .....:                      'd': [1., 2., 3, 4]})
   .....: 

In [368]: data
Out[368]: 
     a    b  c    d
0  bar  one  z  1.0
1  bar  two  y  2.0
2  foo  one  x  3.0
3  foo  two  w  4.0

In [369]: indexed1 = data.set_index('c')

In [370]: indexed1
Out[370]: 
     a    b    d
c               
z  bar  one  1.0
y  bar  two  2.0
x  foo  one  3.0
w  foo  two  4.0

In [371]: indexed2 = data.set_index(['a', 'b'])

In [372]: indexed2
Out[372]: 
         c    d
a   b          
bar one  z  1.0
    two  y  2.0
foo one  x  3.0
    two  w  4.0
```

The `append` keyword option allow you to keep the existing index and append the given columns to a MultiIndex:

```
In [373]: frame = data.set_index('c', drop=False)

In [374]: frame = frame.set_index(['a', 'b'], append=True)

In [375]: frame
Out[375]: 
           c    d
c a   b          
z bar one  z  1.0
y bar two  y  2.0
x foo one  x  3.0
w foo two  w  4.0
```

Other options in `set_index` allow you not drop the index columns.

```
In [376]: data.set_index('c', drop=False)
Out[376]: 
     a    b  c    d
c                  
z  bar  one  z  1.0
y  bar  two  y  2.0
x  foo  one  x  3.0
w  foo  two  w  4.0
```

Reset the index

As a convenience, there is a new function on DataFrame called **`reset_index()`** which transfers the index values into the DataFrame's columns and sets a simple integer index. This is the inverse operation of **`set_index()`**.

```
In [377]: data
Out[377]: 
     a    b  c    d
0  bar  one  z  1.0
1  bar  two  y  2.0
2  foo  one  x  3.0
3  foo  two  w  4.0

In [378]: data.reset_index()
Out[378]: 
   index    a    b  c    d
0      0  bar  one  z  1.0
1      1  bar  two  y  2.0
2      2  foo  one  x  3.0
3      3  foo  two  w  4.0
```

The output is more similar to a SQL table or a record array. The names for the columns derived from the index are the ones stored in the `names` attribute.

You can use the `level` keyword to remove only a portion of the index:

```
In [379]: frame
Out[379]: 
           c    d
c a   b          
z bar one  z  1.0
y bar two  y  2.0
x foo one  x  3.0
w foo two  w  4.0

In [380]: frame.reset_index(level=1)
Out[380]: 
         a  c    d
c b               
z one  bar  z  1.0
y two  bar  y  2.0
x one  foo  x  3.0
w two  foo  w  4.0
```

`reset_index` takes an optional parameter `drop` which if true simply discards the index, instead of putting index values in the DataFrame's columns.

Adding an ad hoc index

You can assign a custom index to the `index` attribute:

```
In [381]: df_idx = pd.DataFrame(range(4))

In [382]: df_idx.index = pd.Index([10, 20, 30, 40], name="a")

In [383]: df_idx
Out[383]: 
    0
a    
10  0
20  1
30  2
40  3
```

Why does assignment fail when using chained indexing?

Copy-on-Write is the new default with pandas 3.0. This means that chained indexing will never work. See this section for more context.

Series Assignment and Index Alignment

When assigning a Series to a DataFrame column, pandas performs automatic alignment based on index labels. This is a fundamental behavior that can be surprising to new users who might expect positional assignment.

Key Points:

- Series values are matched to DataFrame rows by index label
- Position/order in the Series doesn't matter
- Missing index labels result in NaN values
- This behavior is consistent across df[col] = series and df.loc[:, col] = series

Examples: .. ipython:: python

> import pandas as pd
>
> # Create a DataFrame df = pd.DataFrame({‘values’: [1, 2, 3]}, index=[‘x’, ‘y’, ‘z’])
>
> # Series with matching indices (different order) s1 = pd.Series([10, 20, 30], index=[‘z’, ‘x’, ‘y’]) df[‘aligned’] = s1 # Aligns by index, not position print(df)
>
> # Series with partial index match s2 = pd.Series([100, 200], index=[‘x’, ‘z’]) df[‘partial’] = s2 # Missing ‘y’ gets NaN print(df)
>
> # Series with non-matching indices s3 = pd.Series([1000, 2000], index=[‘a’, ‘b’]) df[‘nomatch’] = s3 # All values become NaN print(df)
>
> #Avoiding Confusion: #If you want positional assignment instead of index alignment: # reset the Series index to match DataFrame index df[‘s1_values’] = s1.reindex(df.index)

‹ Previous
PyArrow Functionality

Next
MultiIndex / advanced indexing ›

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